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DTK

DTK 24.04.1
兼容性手册

DTK-240401-003-0606-010

developer.hpccube.com

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版本更新

- DTK-24.04：

  更新操作系统与内核兼容性列表。

- DTK-23.04：

  更新数学库兼容性列表，新增通讯组件兼容性列表。

- DTK-22.10：

  更新操作系统与内核兼容性列表，补充支持的其他操作系统。

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1 操作系统和内核兼容性列表

系统	版本	内核
CentOS	7.6	3.10.0-957.el7.x86_64
CentOS	8.5	4.18.0-348.el8.x86_64
Rocky	8.6	4.18.0-348.el8.x86_64
Rocky	9.2	4.18.0-348.el8.x86_64
Ubuntu	20.04.1	5.4.0-42-generic
Ubuntu	22.04	5.15.0-25-generic
NFS	3.2	3.10.0-957.nfs.5.x86_64
NFS	4.0	4.19.113-14.1.nfs4.x86_64
NFS	4.0-Desktop	5.4.0-49-generic
UOS	1021e	4.19.90-2109.1.0.0108.up2.uel20.x86_64
Kylin	v10 SP2	4.19.90-24.4.v2101.ky10.x86_64
Anolis	8.4	4.19.91-23.4.an8.x86_64
Anolis	8.6	4.19.91-26.an8.x86_64
openEuler	22.03	5.10.0-60.18.0.50.oe2203.x86_64
BCLinux	8.2	4.19.0-240.23.11.el8_2.bclinux.x86_64

2 异构程序兼容性

基于 DTK-21.10、DTK-21.10.1、DTK-22.04、DTK-22.04.1、DTK-22.04.2、DTK-22.10、DTK-22.10.1、DTK-23.04、DTK-23.04.1、DTK-23.10、DTK-23.10.1、DTK-24.04 编译的异构可执行程序和异构共享库可以直接在 DTK-24.04.1 上使用。

3 CUDA API 兼容性列表

3.1 Device Management

CUDA	HIP	CUDA version*

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cudaChooseDevice	hipChooseDevice	
cudaDeviceGetAttribute	hipDeviceGetAttribute	
cudaDeviceGetByPCIBusId	hipDeviceGetByPCIBusId	
cudaDeviceGetCacheConfig	hipDeviceGetCacheConfig	
cudaDeviceGetLimit	hipDeviceGetLimit	
cudaDeviceGetNvSciSyncAttributes		10.2
cudaDeviceGetPCIBusId	hipDeviceGetPCIBusId	
cudaDeviceGetSharedMemConfig	hipDeviceGetSharedMemConfig	
cudaDeviceGetStreamPriorityRange	hipDeviceGetStreamPriorityRange	
cudaDeviceReset	hipDeviceReset	
cudaDeviceSetCacheConfig	hipDeviceSetCacheConfig	
cudaDeviceSetLimit		
cudaDeviceSetSharedMemConfig	hipDeviceSetSharedMemConfig	
cudaDeviceSynchronize	hipDeviceSynchronize	
cudaGetDevice	hipGetDevice	
cudaGetDeviceCount	hipGetDeviceCount	
cudaGetDeviceFlags		
cudaGetDeviceProperties	hipGetDeviceProperties	

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cudaIpcCloseMemHandle	hipIpcCloseMemHandle	
cudaIpcGetEventHandle		
cudaIpcGetMemHandle	hipIpcGetMemHandle	
cudaIpcOpenEventHandle		
cudaIpcOpenMemHandle	hipIpcOpenMemHandle	
cudaSetDevice	hipSetDevice	
cudaSetDeviceFlags	hipSetDeviceFlags	
cudaSetValidDevices		
cudaDeviceGetP2PAttribute		8.0

3.2 Error Handling

CUDA	HIP	CUDA version*
cudaGetErrorName	hipGetErrorName	
cudaGetErrorString	hipGetErrorString	
cudaGetLastError	hipGetLastError	
cudaPeekAtLastError	hipPeekAtLastError	

3.3 Stream Management

CUDA	HIP	CUDA version*
cudaStreamAddCallback	hipStreamAddCallback	
cudaStreamAttachMemAsync		
cudaStreamBeginCapture		10.0
cudaStreamEndCapture		10.0
cudaStreamIsCapturing		10.0
cudaStreamGetCaptureInfo		10.1

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cudaStreamCreate	hipStreamCreate	
cudaStreamCreateWithFlags	hipStreamCreateWithFlags	
cudaStreamCreateWithPriority	hipStreamCreateWithPriority	
cudaStreamDestroy	hipStreamDestroy	
cudaStreamGetFlags	hipStreamGetFlags	
cudaStreamGetPriority	hipStreamGetPriority	
cudaStreamQuery	hipStreamQuery	
cudaStreamSynchronize	hipStreamSynchronize	
cudaStreamWaitEvent	hipStreamWaitEvent	
cudaThreadExchangeStreamCaptureMode		10.1

3.4 Event Management

CUDA	HIP	CUDA version*
cudaEventCreate	hipEventCreate	
cudaEventCreateWithFlags	hipEventCreateWithFlags	
cudaEventDestroy	hipEventDestroy	
cudaEventElapsedTime	hipEventElapsedTime	
cudaEventQuery	hipEventQuery	
cudaEventRecord	hipEventRecord	
cudaEventSynchronize	hipEventSynchronize	

3.5 External Resource Interoperability

CUDA	HIP	CUDA version*
cudaSignalExternalSemaphoresAsync		10.0
cudaWaitExternalSemaphoresAsync		10.0
cudaImportExternalMemory		10.0
cudaExternalMemoryGetMappedBuffer		10.0

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cudaExternalMemoryGetMappedMipmappedArray		10.0
cudaDestroyExternalMemory		10.0
cudaImportExternalSemaphore		10.0
cudaDestroyExternalSemaphore		10.0

3.6 Execution Control

CUDA	HIP	CUDA version*
cudaFuncGetAttributes	hipFuncGetAttributes	
cudaFuncSetAttribute		9.0
cudaFuncSetCacheConfig	hipFuncSetCacheConfig	
cudaFuncSetSharedMemConfig		
cudaGetParameterBuffer		
cudaGetParameterBufferV2		
cudaLaunchKernel	hipLaunchKernel	
cudaSetDoubleForDevice		
cudaSetDoubleForHost		
cudaLaunchCooperativeKernel		9.0
cudaLaunchCooperativeKernel MultiDevice		9.0
cudaLaunchHostFunc		10.0

3.7 Occupancy

CUDA	HIP	CUDA version*
cudaOccupancyMaxActiveBlocksPer Multiprocessor		
cudaOccupancyMaxActiveBlocksPer MultiprocessorWithFlags		

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3.8 Execution Control [DEPRECATED since 7.0]

CUDA	HIP	CUDA version*
cudaConfigureCall	hipConfigureCall	
cudaLaunch	hipLaunchByPtr	
cudaSetupArgument	hipSetupArgument	

3.9 Memory Management

CUDA	HIP	CUDA version*
cudaArrayGetInfo		
cudaFree	hipFree	
cudaFreeArray	hipFreeArray	
cudaFreeHost	hipHostFree	
cudaFreeMipmappedArray		
cudaGetMipmappedArrayLevel		
cudaGetSymbolAddress	hipGetSymbolAddress	
cudaGetSymbolSize	hipGetSymbolSize	
cudaHostAlloc	hipHostMalloc	
cudaHostGetDevicePointer	hipHostGetDevicePointer	
cudaHostGetFlags	hipHostGetFlags	
cudaHostRegister	hipHostRegister	
cudaHostUnregister	hipHostUnregister	
cudaMalloc	hipMalloc	
cudaMalloc3D	hipMalloc3D	
cudaMalloc3DArray	hipMalloc3DArray	
cudaMallocArray	hipMallocArray	
cudaMallocHost	hipHostMalloc	

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cudaMallocManaged	hipMallocManaged	
cudaMallocMipmappedArray		
cudaMallocPitch		
cudaMemGetInfo	hipMemGetInfo	
cudaMemPrefetchAsync		8.0
cudaMemcpy	hipMemcpy	
cudaMemcpy2D	hipMemcpy2D	
cudaMemcpy2DArrayToArray		
cudaMemcpy2DAsync	hipMemcpy2DAsync	
cudaMemcpy2DFromArray		
cudaMemcpy2DFromArrayAsync		
cudaMemcpy2DToArray	hipMemcpy2DToArray	
cudaMemcpy2DToArrayAsync		
cudaMemcpy3D	hipMemcpy3D	
cudaMemcpy3DAsync		
cudaMemcpy3DPeer		
cudaMemcpy3DPeerAsync		
cudaMemcpyArrayToArray		
cudaMemcpyAsync	hipMemcpyAsync	
cudaMemcpyFromArray	hipMemcpyFromArray	
cudaMemcpyFromArrayAsync		
cudaMemcpyFromSymbol	hipMemcpyFromSymbol	
cudaMemcpyFromSymbolAsync	hipMemcpyFromSymbolAsync	
cudaMemcpyPeer	hipMemcpyPeer	
cudaMemcpyPeerAsync	hipMemcpyPeerAsync	
cudaMemcpyToArray	hipMemcpyToArray	
cudaMemcpyToArrayAsync		
cudaMemcpyToSymbol	hipMemcpyToSymbol	

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cudaMemcpyToSymbolAsync	hipMemcpyToSymbolAsync	
cudaMemset	hipMemset	
cudaMemset2D	hipMemset2D	
cudaMemset2DAsync	hipMemset2DAsync	
cudaMemset3D	hipMemset3D	
cudaMemset3DAsync	hipMemset3DAsync	
cudaMemsetAsync	hipMemsetAsync	
make_cudaExtent	make_hipExtent	
make_cudaPitchedPtr		
make_cudaPos	make_hipPos	

3.10 Unified Addressing

CUDA	HIP	CUDA version*
cudaPointerGetAttributes	hipPointerGetAttributes	

3.11 Peer Device Memory Access

CUDA	HIP	CUDA version*
cudaDeviceCanAccessPeer	hipDeviceCanAccessPeer	
cudaDeviceDisablePeerAccess	hipDeviceDisablePeerAccess	
cudaDeviceEnablePeerAccess	hipDeviceEnablePeerAccess	

3.12 Version Management

CUDA	HIP	CUDA version*
cudaDriverGetVersion	hipDriverGetVersion	
cudaRuntimeGetVersion	hipRuntimeGetVersion	

3.13 Graph Management

CUDA	HIP	CUDA version*
cudaGraphAddChildGraphNode		10.0
cudaGraphAddDependencies		10.0
cudaGraphAddEmptyNode		10.0

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cudaGraphAddHostNode		10.0
cudaGraphAddKernelNode		10.0
cudaGraphAddMemcpyNode		10.0
cudaGraphAddMemsetNode		10.0
cudaGraphChildGraphNodeGetGraph		10.0
cudaGraphClone		10.0
cudaGraphCreate		10.0
cudaGraphDestroy		10.0
cudaGraphDestroyNode		10.0
cudaGraphExecDestroy		10.0
cudaGraphGetEdges		10.0
cudaGraphGetNodes		10.0
cudaGraphGetRootNodes		10.0
cudaGraphHostNodeGetParams		10.0
cudaGraphHostNodeSetParams		10.0
cudaGraphInstantiate		10.0
cudaGraphExecKernelNodeSetParams		10.1
cudaGraphExecMemcpyNodeSetParams		10.2
cudaGraphExecMemsetNodeSetParams		10.2
cudaGraphExecHostNodeSetParams		10.2
cudaGraphExecUpdate		10.2
cudaGraphKernelNodeGetParams		10.0
cudaGraphKernelNodeSetParams		10.0
cudaGraphLaunch		10.0
cudaGraphMemcpyNodeGetParams		10.0
cudaGraphMemcpyNodeSetParams		10.0
cudaGraphMemsetNodeGetParams		10.0
cudaGraphMemsetNodeSetParams		10.0

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cudaGraphNodeFindInClone		10.0
cudaGraphNodeGetDependencies		10.0
cudaGraphNodeGetDependentNodes		10.0
cudaGraphNodeGetType		10.0
cudaGraphRemoveDependencies		10.0

3.14 Profiler Control

CUDA	HIP	CUDA version*
cudaProfilerInitialize		
cudaProfilerStart	hipProfilerStart	
cudaProfilerStop	hipProfilerStop	

3.15 Device Library 支持列表

数学函数支持列表

printf	umul64hi	__float2half_rn	__ull2double_rz
__popc	mul64hi	__half2float	__ull2float_rd
__popcll	__umulhi	__float2int_rd	__ull2float_rn
__clz	mulhi	__float2int_rn	__ull2float_ru
__clzll	mulhi	__float2int_ru	__ull2float_rz
__ffs	mulhi	__float2int_rz	float2int
__ffsll	__urhadd	__float2ll_rd	float2uint
__ffs	__usad	__float2ll_rn	int2float
__ffsll	__lane_id	__float2ll_ru	uint2float
__brev	__mbcnt_lo	__float2ll_rz	_clock64
__brevll	__mbcnt_hi	__float2uint_rd	_clock
__lastbit_u32_u64	__hip_ds_bpermute	__float2uint_rn	lock64
__bitextract_u32	__hip_ds_bpermutef	__float2uint_ru	lock
__bitextract_	__hip_ds_permu	__float2uint_rz	__named_sync

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u64	te		
__bitinsert_u32	__hip_ds_permutef	__float2ull_rd	__clock64
__bitinsert_u64	__hip_ds_swizzle_N	__float2ull_rn	__clock
__byte_perm	__hip_ds_swizzlef_N	__float2ull_ru	clock64
__hadd	__hip_move_dpp_N	__float2ull_rz	clock
__mul24	__shfl	__float_as_int	__named_sync
__mul64hi	__shfl_up	float_as_int	__all
__mulhi	__shfl_down	__float_as_uint	__any
__rhadd	__shfl_xor	float_as_uint	__ballot
__sad	__double2float_rd	__hiloint2double	__ballot64
__uhadd	__double2float_rn	__int2double_rn	__lanemask_gt
__umul24	__double2float_ru	__int2float_rd	__lanemask_lt
__umul64hi	__double2float_rz	__int2float_rn	__lanemask_eq
__umulhi	__double2hiint	__int2float_ru	__local_to_generic
__urhadd	__double2loint	__int2float_rz	__get_dynamicgroupbaseptr
__usad	__double2int_rd	__int_as_float	__get_dynamicgroupbaseptr
__byte_perm	__double2int_rn	int_as_float	__amdgcn_get_dynamicgroupbaseptr

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__hadd	__double2int_ru	__ll2double_rd	__threadfence
__mul24	__double2int_rz	__ll2double_rn	__threadfence_block
mul24	__double2ll_rd	__ll2double_ru	__threadfence_system
__mul64hi	__double2ll_rn	__ll2double_rz	abort
mul64hi	__double2ll_ru	__ll2float_rd	_wassert
__mulhi	__double2ll_rz	__ll2float_rn	__assert_fail
mulhi	__double2uint_rd	__ll2float_ru	__assertfail
__rhadd	__double2uint_rn	__ll2float_rz	__work_group_barrier
__sad	__double2uint_ru	__longlong_as_double	__barrier
__uhadd	__double2uint_rz	__uint2double_rn	__syncthreads
__umul24	__double2ull_rd	__uint2float_rd	__syncthreads_count
umul24	__double2ull_rn	__uint2float_rn	__syncthreads_and
__umul64hi	__double2ull_ru	__uint2float_ru	__syncthreads_or
mul64hi	__double2ull_rz	__uint2float_rz	__smid
umul64hi	__double_as_longlong	__uint_as_float	__hip_memcpy
umul64hi	__float2half_rn	__ull2double_rd	__hip_memset
mul64hi	__half2float	__ull2double_rn	memcpy
	__half	__ull2double_ru	memset

原子函数支持列表

atomicCAS	atomicMin
atomicCAS_system	atomicMin_system
atomicAdd	atomicOr
atomicAdd_system	atomicOr_system

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atomicAddNoRet	atomicXor
atomicSub	atomicXor_system
atomicSub_system	atomicInc
atomicExch	atomicDec
atomicExch_system	atomicAnd
	atomicAnd_system

4 CUDA API 支持列表

HIP 系统提供了对 CUDA 函数的直接支持，以下 CUDA 函数可以直接使用。

cudaDeviceReset	cudaGetMipmappedArrayLevel
cudaDeviceSetCacheConfig	cudaMemcpy3D
cudaDeviceSetSharedMemConfig	cudaArrayGetInfo
cudaDeviceGetByPCIBusId	cudaMemcpy
cudaDeviceGetPCIBusId	cudaMemcpyPeer
cudaIpcGetEventHandle	cudaMemcpyToArray
cudaIpcOpenEventHandle	cudaMemcpyFromArray
cudaIpcGetMemHandle	cudaMemcpy2D
cudaIpcOpenMemHandle	cudaMemcpy2DToArray
cudaIpcCloseMemHandle	cudaMemcpy2DFromArray
cudaChooseDevice	cudaMemcpyToSymbol
cudaSetDevice	cudaMemcpyFromSymbol
cudaSetDeviceFlags	cudaMemcpyPeerAsync
cudaGetDeviceFlags	cudaMemcpyToSymbolAsync
cudaStreamCreate	cudaMemcpyFromSymbolAsync
cudaStreamAddCallback	cudaMemset
cudaStreamSynchronize	cudaMemset2D
cudaStreamQuery	cudaMemset3D
cudaEventCreate	cudaGetSymbolAddress
cudaEventQuery	cudaGetSymbolSize

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cudaEventSynchronize	cudaMemPrefetchAsync
cudaEventElapsedTime	cudaMemAdvise
cudaLaunchKernel	cudaMemRangeGetAttribute
cudaLaunchCooperativeKernel	cudaMemRangeGetAttributes
cudaLaunchCooperativeKernelMultiDevice	cudaPointerGetAttributes
cudaFuncSetCacheConfig	cudaDeviceCanAccessPeer
cudaConfigureCall	cudaDeviceEnablePeerAccess
cudaSetupArgument	cudaDeviceDisablePeerAccess
cudaMallocHost	cudaGetChannelDesc
cudaMallocPitch	cudaBindTexture
cudaMallocArray	cudaBindTexture2D
cudaFreeHost	cudaBindTextureToArray
cudaFreeArray	cudaBindTextureToMipmappedArray
cudaFreeMipmappedArray	cudaUnbindTexture
cudaHostAlloc	cudaGetTextureAlignmentOffset
cudaHostRegister	cudaGetTextureReference
cudaHostUnregister	cudaCreateTextureObject
cudaHostGetDevicePointer	cudaDestroyTextureObject
cudaHostGetFlags	cudaGetTextureObjectResourceDesc
cudaMalloc3D	cudaGetTextureObjectTextureDesc
cudaMalloc3DArray	cudaGetTextureObjectResourceViewDesc
cudaMallocMipmappedArray	cudaDriverGetVersion
cudaDeviceReset	cudaGetMipmappedArrayLevel

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5 数学库兼容性列表

5.1 BLAS

5.2 BLAS Data types

type	CUDA	HIP	CUDA version*
define	CUBLAS_VER_MAJOR		10.1 Update 2
define	CUBLAS_VER_MINOR		10.1 Update 2
define	CUBLAS_VER_PATCH		10.1 Update 2
define	CUBLAS_VER_BUILD		10.1 Update 2
define	CUBLAS_VERSION		10.1 Update 2
enum	cublasStatus_t	hipblasStatus_t	
0	CUBLAS_STATUS_SUCCESS	HIPBLAS_STATUS_SUCCESS	
1	CUBLAS_STATUS_NOT_INITIALIZED	HIPBLAS_STATUS_NOT_INITIALIZED	
3	CUBLAS_STATUS_ALLOC_FAILED	HIPBLAS_STATUS_ALLOC_FAILED	
7	CUBLAS_STATUS_INVALID_VALUE	HIPBLAS_STATUS_INVALID_VALUE	
8	CUBLAS_STATUS_ARCH_MISMATCH	HIPBLAS_STATUS_ARCH_MISMATCH	
11	CUBLAS_STATUS_MAPPING_ERROR	HIPBLAS_STATUS_MAPPING_ERROR	
13	CUBLAS_STATUS_E	HIPBLAS_STATUS_	

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	XECUTION_FAILED	EXECUTION_FAILED	
14	CUBLAS_STATUS_INTERNAL_ERROR	HIPBLAS_STATUS_INTERNAL_ERROR	
15	CUBLAS_STATUS_NOT_SUPPORTED	HIPBLAS_STATUS_NOT_SUPPORTED	
16	CUBLAS_STATUS_LICENSE_ERROR		
enum	cublasOperation_t	hipblasOperation_t	
0	CUBLAS_OP_N	HIPBLAS_OP_N	
1	CUBLAS_OP_T	HIPBLAS_OP_T	
2	CUBLAS_OP_C	HIPBLAS_OP_C	
2	CUBLAS_OP_HERMITAN	HIPBLAS_OP_C	10.1
3	CUBLAS_OP_CONJG		10.1
enum	cublasFillMode_t	hipblasFillMode_t	
0	CUBLAS_FILL_MODE_LOWER	HIPBLAS_FILL_MODE_LOWER	
1	CUBLAS_FILL_MODE_UPPER	HIPBLAS_FILL_MODE_UPPER	
2	CUBLAS_FILL_MODE_FULL	HIPBLAS_FILL_MODE_FULL	10.1
enum	cublasDiagType_t	hipblasDiagType_t	
0	CUBLAS_DIAG_NON_UNIT	HIPBLAS_DIAG_NON_UNIT	
1	CUBLAS_DIAG_UNIT	HIPBLAS_DIAG_UNIT	

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enum	cublasSideMode_t	hipblasSideMode_t	
0	CUBLAS_SIDE_LEFT	HIPBLAS_SIDE_LEFT	
1	CUBLAS_SIDE_RIGHT	HIPBLAS_SIDE_RIGHT	
enum	cublasPointerMode_t	hipblasPointerMode_t	
0	CUBLAS_POINTER_MODE_HOST	HIPBLAS_POINTER_MODE_HOST	
1	CUBLAS_POINTER_MODE_DEVICE	HIPBLAS_POINTER_MODE_DEVICE	
enum	cublasAtomicsMode_t	hipblasAtomicsMode_t	
0	CUBLAS_ATOMICS_NOT_ALLOWED	HIPBLAS_ATOMICS_NOT_ALLOWED	
1	CUBLAS_ATOMICS_ALLOWED	HIPBLAS_ATOMICS_ALLOWED	
enum	cublasGemmAlgo_t	hipblasGemmAlgo_t	8
-1	CUBLAS_GEMM_DFALT	HIPBLAS_GEMM_DEFAULT	8
-1	CUBLAS_GEMM_DEFAULT	HIPBLAS_GEMM_DEFAULT	8
0	CUBLAS_GEMM_ALGO0		8
1	CUBLAS_GEMM_ALGO1		8
2	CUBLAS_GEMM_ALGO2		8
3	CUBLAS_GEMM_ALGO3		8

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4	CUBLAS_GEMM_ALGO4		8
5	CUBLAS_GEMM_ALGO5		8
6	CUBLAS_GEMM_ALGO6		8
7	CUBLAS_GEMM_ALGO7		8
8	CUBLAS_GEMM_ALGO8		9
9	CUBLAS_GEMM_ALGO9		9
10	CUBLAS_GEMM_ALGO10		9
11	CUBLAS_GEMM_ALGO11		9
12	CUBLAS_GEMM_ALGO12		9
13	CUBLAS_GEMM_ALGO13		9
14	CUBLAS_GEMM_ALGO14		9
15	CUBLAS_GEMM_ALGO15		9
16	CUBLAS_GEMM_ALGO16		9
17	CUBLAS_GEMM_ALGO17		9
18	CUBLAS_GEMM_AL		9.2

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	GO18		
19	CUBLAS_GEMM_ALGO19		9.2
20	CUBLAS_GEMM_ALGO20		9.2
21	CUBLAS_GEMM_ALGO21		9.2
22	CUBLAS_GEMM_ALGO22		9.2
23	CUBLAS_GEMM_ALGO23		9.2
99	CUBLAS_GEMM_DEFAULT_TENSOR_OP		9
99	CUBLAS_GEMM_DFALT_TENSOR_OP		9
100	CUBLAS_GEMM_ALGO0_TENSOR_OP		9
101	CUBLAS_GEMM_ALGO1_TENSOR_OP		9
102	CUBLAS_GEMM_ALGO2_TENSOR_OP		9
103	CUBLAS_GEMM_ALGO3_TENSOR_OP		9
104	CUBLAS_GEMM_ALGO4_TENSOR_OP		9
105	CUBLAS_GEMM_ALGO5_TENSOR_OP		9.2
106	CUBLAS_GEMM_ALGO6_TENSOR_OP		9.2

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107	CUBLAS_GEMM_ALGO7_TENSOR_OP		9.2
108	CUBLAS_GEMM_ALGO8_TENSOR_OP		9.2
109	CUBLAS_GEMM_ALGO9_TENSOR_OP		9.2
110	CUBLAS_GEMM_ALGO10_TENSOR_OP		9.2
111	CUBLAS_GEMM_ALGO11_TENSOR_OP		9.2
112	CUBLAS_GEMM_ALGO12_TENSOR_OP		9.2
113	CUBLAS_GEMM_ALGO13_TENSOR_OP		9.2
114	CUBLAS_GEMM_ALGO14_TENSOR_OP		9.2
115	CUBLAS_GEMM_ALGO15_TENSOR_OP		9.2
enum	cublasMath_t		9
0	CUBLAS_DEFAULT_MATH		9
1	CUBLAS_TENSOR_OP_MATH		9
enum*	cublasDataType_t		7.5
struct	cublasContext		
struct*	cublasHandle_t	hipblasHandle_t	

5.2.1 BLAS Level-1 Function Reference

CUDA	HIP	CUDA version*

开发者社区

cublasCaxpy	hipblasCaxpy	
cublasCaxpy_v2	hipblasCaxpy	
cublasCcopy	hipblasCcopy	
cublasCcopy_v2	hipblasCcopy	
cublasCdotc	hipblasCdotc	
cublasCdotc_v2	hipblasCdotc	
cublasCdotu	hipblasCdotu	
cublasCdotu_v2	hipblasCdotu	
cublasCrot	hipblasCrot	
cublasCrot_v2	hipblasCrot	
cublasCrotg	hipblasCrotg	
cublasCrotg_v2	hipblasCrotg	
cublasCscal	hipblasCscal	
cublasCscal_v2	hipblasCscal	
cublasCsrot	hipblasCsrot	
cublasCsrot_v2	hipblasCsrot	
cublasCsscal	hipblasCsscal	
cublasCsscal_v2	hipblasCsscal	
cublasCswap	hipblasCswap	
cublasCswap_v2	hipblasCswap	
cublasDasum	hipblasDasum	
cublasDasum_v2	hipblasDasum	
cublasDaxpy	hipblasDaxpy	
cublasDaxpy_v2	hipblasDaxpy	
cublasDcopy	hipblasDcopy	
cublasDcopy_v2	hipblasDcopy	
cublasDdot	hipblasDdot	
cublasDdot_v2	hipblasDdot	

cublasDnrm2	hipblasDnrm2	
cublasDnrm2_v2	hipblasDnrm2	
cublasDrot	hipblasDrot	
cublasDrot_v2	hipblasDrot	
cublasDrotg	hipblasDrotg	
cublasDrotg_v2	hipblasDrotg	
cublasDrotm	hipblasDrotm	
cublasDrotm_v2	hipblasDrotm	
cublasDrotmg	hipblasDrotmg	
cublasDrotmg_v2	hipblasDrotmg	
cublasDscal	hipblasDscal	
cublasDscal_v2	hipblasDscal	
cublasDswap	hipblasDswap	
cublasDswap_v2	hipblasDswap	
cublasDzasum	hipblasDzasum	
cublasDzasum_v2	hipblasDzasum	
cublasDznrm2	hipblasDznrm2	
cublasDznrm2_v2	hipblasDznrm2	
cublasIcamax	hipblasIcamax	
cublasIcamax_v2	hipblasIcamax	
cublasIcamin	hipblasIcamin	
cublasIcamin_v2	hipblasIcamin	
cublasIdamax	hipblasIdamax	
cublasIdamax_v2	hipblasIdamax	
cublasIdamin	hipblasIdamin	
cublasIdamin_v2	hipblasIdamin	
cublasIsamax	hipblasIsamax	
cublasIsamax_v2	hipblasIsamax	

cublasIsamin	hipblasIsamin	
cublasIsamin_v2	hipblasIsamin	
cublasIzamax	hipblasIzamax	
cublasIzamax_v2	hipblasIzamax	
cublasIzamin	hipblasIzamin	
cublasIzamin_v2	hipblasIzamin	
cublasNrm2Ex	hipblasNrm2Ex	8
cublasSasum	hipblasSasum	
cublasSasum_v2	hipblasSasum	
cublasSaxpy	hipblasSaxpy	
cublasSaxpy_v2	hipblasSaxpy	
cublasScasum	hipblasScasum	
cublasScasum_v2	hipblasScasum	
cublasScnrm2	hipblasScnrm2	
cublasScnrm2_v2	hipblasScnrm2	
cublasScopy	hipblasScopy	
cublasScopy_v2	hipblasScopy	
cublasSdot	hipblasSdot	
cublasSdot_v2	hipblasSdot	
cublasSnrm2	hipblasSnrm2	
cublasSnrm2_v2	hipblasSnrm2	
cublasSrot	hipblasSrot	
cublasSrot_v2	hipblasSrot	
cublasSrotg	hipblasSrotg	
cublasSrotg_v2	hipblasSrotg	
cublasSrotm	hipblasSrotm	
cublasSrotm_v2	hipblasSrotm	
cublasSrotmg	hipblasSrotmg	

cublasSrotmg_v2	hipblasSrotmg	
cublasSscal	hipblasSscal	
cublasSscal_v2	hipblasSscal	
cublasSswap	hipblasSswap	
cublasSswap_v2	hipblasSswap	
cublasZaxpy	hipblasZaxpy	
cublasZaxpy_v2	hipblasZaxpy	
cublasZcopy	hipblasZcopy	
cublasZcopy_v2	hipblasZcopy	
cublasZdotc	hipblasZdotc	
cublasZdotc_v2	hipblasZdotc	
cublasZdotu	hipblasZdotu	
cublasZdotu_v2	hipblasZdotu	
cublasZdrot	hipblasZdrot	
cublasZdrot_v2	hipblasZdrot	
cublasZdscal	hipblasZdscal	
cublasZdscal_v2	hipblasZdscal	
cublasZrot	hipblasZrot	
cublasZrot_v2	hipblasZrot	
cublasZrotg	hipblasZrotg	
cublasZrotg_v2	hipblasZrotg	
cublasZscal	hipblasZscal	
cublasZscal_v2	hipblasZscal	
cublasZswap	hipblasZswap	
cublasZswap_v2	hipblasZswap	

5.2.2 BLAS Level-2 Function Reference

CUDA	HIP	CUDA version*
cublasCgbmv	hipblasCgbmv	

cublasCgbmv_v2	hipblasCgbmv	
cublasCgemv	hipblasCgemv	
cublasCgemv_v2	hipblasCgemv	
cublasCgerc	hipblasCgerc	
cublasCgerc_v2	hipblasCgerc	
cublasCgeru	hipblasCgeru	
cublasCgeru_v2	hipblasCgeru	
cublasChbmv	hipblasChbmv	
cublasChbmv_v2	hipblasChbmv	
cublasChemv	hipblasChemv	
cublasChemv_v2	hipblasChemv	
cublasCher	hipblasCher	
cublasCher2	hipblasCher2	
cublasCher2_v2	hipblasCher2	
cublasCher_v2	hipblasCher	
cublasChpmv	hipblasChpmv	
cublasChpmv_v2	hipblasChpmv	
cublasChpr	hipblasChpr	
cublasChpr2	hipblasChpr2	
cublasChpr2_v2	hipblasChpr2	
cublasChpr_v2	hipblasChpr	
cublasCsymv	hipblasCsymv	
cublasCsymv_v2	hipblasCsymv	
cublasCsyr	hipblasCsyr	
cublasCsyr2	hipblasCsyr2	
cublasCsyr2_v2	hipblasCsyr2	
cublasCsyr_v2	hipblasCsyr	
cublasCtbmv	hipblasCtbmv	

cublasCtbmv_v2	hipblasCtbmv	
cublasCtbsv	hipblasCtbsv	
cublasCtbsv_v2	hipblasCtbsv	
cublasCtpmv	hipblasCtpmv	
cublasCtpmv_v2	hipblasCtpmv	
cublasCtpsv	hipblasCtpsv	
cublasCtpsv_v2	hipblasCtpsv	
cublasCtrmv	hipblasCtrmv	
cublasCtrmv_v2	hipblasCtrmv	
cublasCtrsv	hipblasCtrsv	
cublasCtrsv_v2	hipblasCtrsv	
cublasDgbmv	hipblasDgbmv	
cublasDgbmv_v2	hipblasDgbmv	
cublasDgemv	hipblasDgemv	
cublasDgemv_v2	hipblasDgemv	
cublasDger	hipblasDger	
cublasDger_v2	hipblasDger	
cublasDsbmv	hipblasDsbmv	
cublasDsbmv_v2	hipblasDsbmv	
cublasDspmv	hipblasDspmv	
cublasDspmv_v2	hipblasDspmv	
cublasDspr	hipblasDspr	
cublasDspr2	hipblasDspr2	
cublasDspr2_v2	hipblasDspr2	
cublasDspr_v2	hipblasDspr	
cublasDsymv	hipblasDsymv	
cublasDsymv_v2	hipblasDsymv	
cublasDsyr	hipblasDsyr	

cublasDsyr2	hipblasDsyr2	
cublasDsyr2_v2	hipblasDsyr2	
cublasDsyr_v2	hipblasDsyr	
cublasDtbmv	hipblasDtbmv	
cublasDtbmv_v2	hipblasDtbmv	
cublasDtbsv	hipblasDtbsv	
cublasDtbsv_v2	hipblasDtbsv	
cublasDtpmv	hipblasDtpmv	
cublasDtpmv_v2	hipblasDtpmv	
cublasDtpsv	hipblasDtpsv	
cublasDtpsv_v2	hipblasDtpsv	
cublasDtrmv	hipblasDtrmv	
cublasDtrmv_v2	hipblasDtrmv	
cublasDtrsv	hipblasDtrsv	
cublasDtrsv_v2	hipblasDtrsv	
cublasSgbmv	hipblasSgbmv	
cublasSgbmv_v2	hipblasSgbmv	
cublasSgemv	hipblasSgemv	
cublasSgemv_v2	hipblasSgemv	
cublasSger	hipblasSger	
cublasSger_v2	hipblasSger	
cublasSsbmv	hipblasSsbmv	
cublasSsbmv_v2	hipblasSsbmv	
cublasSspmv	hipblasSspmv	
cublasSspmv_v2	hipblasSspmv	
cublasSspr	hipblasSspr	
cublasSspr2	hipblasSspr2	
cublasSspr2_v2	hipblasSspr2	

cublasSspr_v2	hipblasSspr	
cublasSsymv	hipblasSsymv	
cublasSsymv_v2	hipblasSsymv	
cublasSsyr	hipblasSsyr	
cublasSsyr2	hipblasSsyr2	
cublasSsyr2_v2	hipblasSsyr2	
cublasSsyr_v2	hipblasSsyr	
cublasStbmv	hipblasStbmv	
cublasStbmv_v2	hipblasStbmv	
cublasStbsv	hipblasStbsv	
cublasStbsv_v2	hipblasStbsv	
cublasStpmv	hipblasStpmv	
cublasStpmv_v2	hipblasStpmv	
cublasStpsv	hipblasStpsv	
cublasStpsv_v2	hipblasStpsv	
cublasStrmv	hipblasStrmv	
cublasStrmv_v2	hipblasStrmv	
cublasStrsv	hipblasStrsv	
cublasStrsv_v2	hipblasStrsv	
cublasZgbmv	hipblasZgbmv	
cublasZgbmv_v2	hipblasZgbmv	
cublasZgemv	hipblasZgemv	
cublasZgemv_v2	hipblasZgemv	
cublasZgerc	hipblasZgerc	
cublasZgerc_v2	hipblasZgerc	
cublasZgeru	hipblasZgeru	
cublasZgeru_v2	hipblasZgeru	
cublasZhbmv	hipblasZhbmv	

cublasZhbmv_v2	hipblasZhbmv	
cublasZhemv	hipblasZhemv	
cublasZhemv_v2	hipblasZhemv	
cublasZher	hipblasZher	
cublasZher2	hipblasZher2	
cublasZher2_v2	hipblasZher2	
cublasZher_v2	hipblasZher	
cublasZhpmv	hipblasZhpmv	
cublasZhpmv_v2	hipblasZhpmv	
cublasZhpr	hipblasZhpr	
cublasZhpr2	hipblasZhpr2	
cublasZhpr2_v2	hipblasZhpr2	
cublasZhpr_v2	hipblasZhpr	
cublasZsymv	hipblasZsymv	
cublasZsymv_v2	hipblasZsymv	
cublasZsyr	hipblasZsyr	
cublasZsyr2	hipblasZsyr2	
cublasZsyr2_v2	hipblasZsyr2	
cublasZsyr_v2	hipblasZsyr	
cublasZtbmv	hipblasZtbmv	
cublasZtbmv_v2	hipblasZtbmv	
cublasZtbsv	hipblasZtbsv	
cublasZtbsv_v2	hipblasZtbsv	
cublasZtpmv	hipblasZtpmv	
cublasZtpmv_v2	hipblasZtpmv	
cublasZtpsv	hipblasZtpsv	
cublasZtpsv_v2	hipblasZtpsv	
cublasZtrmv	hipblasZtrmv	

cublasZtrmv_v2	hipblasZtrmv	
cublasZtrsv	hipblasZtrsv	
cublasZtrsv_v2	hipblasZtrsv	

5.2.3 BLAS Level-3 Function Reference

CUDA	HIP	CUDA version*
cublasCgemm	hipblasCgemm	
cublasCgemm3m		8
cublasCgemm3mBatched		8
cublasCgemm3mEx		8
cublasCgemm3mStrided Batched		8
cublasCgemmBatched	hipblasCgemmBatched	
cublasCgemmStridedBat ched	hipblasCgemmStridedBatc hed	8
cublasCgemm_v2	hipblasCgemm	
cublasChemm	hipblasChemm	
cublasChemm_v2	hipblasChemm	
cublasCher2k	hipblasCher2k	
cublasCher2k_v2	hipblasCher2k	
cublasCherk	hipblasCherk	
cublasCherk_v2	hipblasCherk	
cublasCherkx	hipblasCherkx	
cublasCsymm	hipblasCsymm	
cublasCsymm_v2	hipblasCsymm	
cublasCsyr2k	hipblasCsyr2k	
cublasCsyr2k_v2	hipblasCsyr2k	
cublasCsyrk	hipblasCsyrk	
cublasCsyrk_v2	hipblasCsyrk	

cublasCsyrkx	hipblasCsyrkx	
cublasCtrmm	hipblasCtrmm	
cublasCtrmm_v2	hipblasCtrmm	
cublasCtrsm	hipblasCtrsm	
cublasCtrsm_v2	hipblasCtrsm	
cublasDgemm	hipblasDgemm	
cublasDgemmBatched	hipblasDgemmBatched	
cublasDgemmStridedBatched	hipblasDgemmStridedBatched	8
cublasDgemm_v2	hipblasDgemm	
cublasDsymm	hipblasDsymm	
cublasDsymm_v2	hipblasDsymm	
cublasDsyr2k	hipblasDsyr2k	
cublasDsyr2k_v2	hipblasDsyr2k	
cublasDsyrk	hipblasDsyrk	
cublasDsyrk_v2	hipblasDsyrk	
cublasDsyrkx	hipblasDsyrkx	
cublasDtrmm	hipblasDtrmm	
cublasDtrmm_v2	hipblasDtrmm	
cublasDtrsm	hipblasDtrsm	
cublasDtrsm_v2	hipblasDtrsm	
cublasHgemm	hipblasHgemm	7.5
cublasHgemmBatched	hipblasHgemmBatched	9
cublasHgemmStridedBatched	hipblasHgemmStridedBatched	8
cublasSgemm	hipblasSgemm	
cublasSgemmBatched	hipblasSgemmBatched	
cublasSgemmStridedBatc	hipblasSgemmStridedBatc	8

hed	hed	
cublasSgemm_v2	hipblasSgemm	
cublasSsymm	hipblasSsymm	
cublasSsymm_v2	hipblasSsymm	
cublasSsyr2k	hipblasSsyr2k	
cublasSsyr2k_v2	hipblasSsyr2k	
cublasSsyrk	hipblasSsyrk	
cublasSsyrk_v2	hipblasSsyrk	
cublasSsyrkx	hipblasSsyrkx	
cublasStrmm	hipblasStrmm	
cublasStrmm_v2	hipblasStrmm	
cublasStrsm	hipblasStrsm	
cublasStrsm_v2	hipblasStrsm	
cublasZgemm	hipblasZgemm	
cublasZgemm3m		8
cublasZgemmBatched	hipblasZgemmBatched	
cublasZgemmStridedBatched	hipblasZgemmStridedBatched	8
cublasZgemm_v2	hipblasZgemm	
cublasZhemm	hipblasZhemm	
cublasZhemm_v2	hipblasZhemm	
cublasZher2k	hipblasZher2k	
cublasZher2k_v2	hipblasZher2k	
cublasZherk	hipblasZherk	
cublasZherk_v2	hipblasZherk	
cublasZherkx	hipblasZherkx	
cublasZsymm	hipblasZsymm	
cublasZsymm_v2	hipblasZsymm	

cublasZsyr2k	hipblasZsyr2k	
cublasZsyr2k_v2	hipblasZsyr2k	
cublasZsyrk	hipblasZsyrk	
cublasZsyrk_v2	hipblasZsyrk	
cublasZsyrkx	hipblasZsyrkx	
cublasZtrmm	hipblasZtrmm	
cublasZtrmm_v2	hipblasZtrmm	
cublasZtrsm	hipblasZtrsm	
cublasZtrsm_v2	hipblasZtrsm	

5.2.4 BLAS Helper Function Referenc

CUDA	HIP	CUDA version*
cublasAlloc		
cublasCreate	hipblasCreate	
cublasCreate_v2	hipblasCreate	
cublasDestroy	hipblasDestroy	
cublasDestroy_v2	hipblasDestroy	
cublasFree		
cublasGetAtomicsMode	hipblasGetAtomicsMode	
cublasGetCudartVersion		10.1
cublasGetError		
cublasGetLoggerCallback		9.2
cublasGetMathMode		9
cublasGetMatrix	hipblasGetMatrix	
cublasGetMatrixAsync	hipblasGetMatrixAsync	
cublasGetPointerMode	hipblasGetPointerMode	
cublasGetPointerMode_v2	hipblasGetPointerMode	

cublasGetProperty		
cublasGetSmCountTarget		11.3
cublasGetStatusName		11.4
cublasGetStatusString		11.4
cublasGetStream	hipblasGetStream	
cublasGetStream_v2	hipblasGetStream	
cublasGetVector	hipblasGetVector	
cublasGetVectorAsync	hipblasGetVectorAsync	
cublasGetVersion		
cublasGetVersion_v2		
cublasInit		
cublasLogCallback		9.2
cublasLoggerConfigure		9.2
cublasMigrateComputeType		11
cublasSetAtomicsMode	hipblasSetAtomicsMode	
cublasSetKernelStream		
cublasSetLoggerCallback		9.2
cublasSetMathMode		
cublasSetMatrix	hipblasSetMatrix	
cublasSetMatrixAsync	hipblasSetMatrixAsync	
cublasSetPointerMode	hipblasSetPointerMode	
cublasSetPointerMode_v2	hipblasSetPointerMode	
cublasSetSmCountTarget		11.3
cublasSetStream	hipblasSetStream	
cublasSetStream_v2	hipblasSetStream	
cublasSetVector	hipblasSetVector	

cublasSetVectorAsync	hipblasSetVectorAsync	
cublasShutdown		
cublasXerbla		

5.2.5 BLAS-like Extension

CUDA	HIP	CUDA version*
cublasAsumEx		10.1
cublasAxpyEx	hipblasAxpyEx	8
cublasCdgmm	hipblasCdgmm	
cublasCgeam	hipblasCgeam	
cublasCgelsBatched	hipblasCgelsBatched	
cublasCgemmEx		8
cublasCgeqrfBatched	hipblasCgeqrfBatched	
cublasCgetrfBatched	hipblasCgetrfBatched	
cublasCgetriBatched	hipblasCgetriBatched	
cublasCgetrsBatched	hipblasCgetrsBatched	
cublasCherk3mEx		8
cublasCherkEx		8
cublasCmatinvBatched		
cublasCopyEx		10.1
cublasCsyrk3mEx		8
cublasCsyrkEx		8
cublasCtpttr		
cublasCtrsmBatched	hipblasCtrsmBatched	
cublasCtrttp		
cublasDdgmm	hipblasDdgmm	
cublasDgeam	hipblasDgeam	
cublasDgelsBatched	hipblasDgelsBatched	
cublasDgeqrfBatched	hipblasDgeqrfBatched	

cublasDgetrfBatched	hipblasDgetrfBatched	
cublasDgetriBatched	hipblasDgetriBatched	
cublasDgetrsBatched	hipblasDgetrsBatched	
cublasDmatinvBatched		
cublasDotEx	hipblasDotEx	8
cublasDotcEx	hipblasDotcEx	8
cublasDtpttr		
cublasDtrsmBatched	hipblasDtrsmBatched	
cublasDtrttp		
cublasGemmBatchedEx	hipblasGemmBatchedEx	9.1
cublasGemmEx	hipblasGemmEx	8
cublasGemmStridedBatchedEx	hipblasGemmStridedBatchedEx	9.1
cublasIamaxEx		10.1
cublasIaminEx		10.1
cublasRotEx	hipblasRotEx	10.1
cublasRotgEx		10.1
cublasRotmEx		10.1
cublasRotmgEx		10.1
cublasScalEx	hipblasScalEx	8
cublasSdgmm	hipblasSdgmm	
cublasSgeam	hipblasSgeam	
cublasSgelsBatched	hipblasSgelsBatched	
cublasSgemmEx		7.5
cublasSgeqrfBatched	hipblasSgeqrfBatched	
cublasSgetrfBatched	hipblasSgetrfBatched	
cublasSgetriBatched	hipblasSgetriBatched	
cublasSgetrsBatched	hipblasSgetrsBatched	

cublasSmatinvBatched		
cublasStpttr		
cublasStrsmBatched	hipblasStrsmBatched	
cublasStrttp		
cublasSwapEx		10.1
cublasUint8gemmBias		8
cublasZdgmm	hipblasZdgmm	
cublasZgeam	hipblasZgeam	
cublasZgelsBatched	hipblasZgelsBatched	
cublasZgeqrfBatched	hipblasZgeqrfBatched	
cublasZgetrfBatched	hipblasZgetrfBatched	
cublasZgetriBatched	hipblasZgetriBatched	
cublasZgetrsBatched	hipblasZgetrsBatched	
cublasZmatinvBatched		
cublasZtpttr		
cublasZtrsmBatched	hipblasZtrsmBatched	
cublasZtrttp		

5.3 MIOpen

CUDA	HIP	CUDA version *
cudnnActivationBackward	hipdnnActivationBackward	
cudnnActivationForward	hipdnnActivationForward	
cudnnAddTensor	hipdnnAddTensor	
cudnnAdvInferVersionCheck		8.0.1
cudnnAdvTrainVersionCheck		8.0.1
cudnnBackendCreateDescriptor		8.0.1
cudnnBackendDestroyDescriptor		8.0.1

cudnnBackendExecute		8.0.1
cudnnBackendFinalize		8.0.1
cudnnBackendGetAttribute		8.0.1
cudnnBackendInitialize		8.0.1
cudnnBackendSetAttribute		8.0.1
cudnnBatchNormalizationBackward	hipdnnBatchNormalizationBackward	
cudnnBatchNormalizationBackwardEx		7.4.1
cudnnBatchNormalizationForwardInference	hipdnnBatchNormalizationForwardInference	
cudnnBatchNormalizationForwardTraining	hipdnnBatchNormalizationForwardTraining	
cudnnBatchNormalizationForwardTrainingEx		7.4.1
cudnnBuildRNNDynamic		8.0.1
cudnnCTCLoss		7.0.5
cudnnCTCLoss_v8		8.0.1
cudnnCnnInferVersionCheck		8.0.2
cudnnCnnTrainVersionCheck		8.0.2
cudnnConvolutionBackwardBias	hipdnnConvolutionBackwardBias	
cudnnConvolutionBackwardData	hipdnnConvolutionBackwardData	
cudnnConvolutionBackwardFilter	hipdnnConvolutionBackwardFilter	
cudnnConvolutionBiasActivationForward		6.0.0

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cudnnConvolutionForward	hipdnnConvolutionForward	
cudnnCreate	hipdnnCreate	
cudnnCreateActivationDescriptor	hipdnnCreateActivationDescriptor	
cudnnCreateAttnDescriptor		7.5.0
cudnnCreateCTCLossDescriptor		7.0.5
cudnnCreateConvolutionDescriptor	hipdnnCreateConvolutionDescriptor	
cudnnCreateDropoutDescriptor	hipdnnCreateDropoutDescriptor	
cudnnCreateFilterDescriptor	hipdnnCreateFilterDescriptor	
cudnnCreateFusedOpsConstParamPack		7.6.0
cudnnCreateFusedOpsPlan		7.6.0
cudnnCreateFusedOpsVariantParamPack		7.6.0
cudnnCreateLRNDescriptor	hipdnnCreateLRNDescriptor	
cudnnCreateOpTensorDescriptor	hipdnnCreateOpTensorDescriptor	
cudnnCreatePoolingDescriptor	hipdnnCreatePoolingDescriptor	
cudnnCreateRNNDataDescriptor		7.2.1
cudnnCreateRNNDescriptor	hipdnnCreateRNNDescriptor	
cudnnCreateReduceTensorDescriptor	hipdnnCreateReduceTensorDescriptor	
cudnnCreateSeqDataDescriptor		7.5.0
cudnnCreateSpatialTransformerDescriptor		5.0.0

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cudnnCreateTensorDescriptor	hipdnnCreateTensorDescriptor	
cudnnCreateTensorTransformDescriptor		7.5.0
cudnnDeriveBNTensorDescriptor	hipdnnDeriveBNTensorDescriptor	
cudnnDeriveNormTensorDescriptor		8.0.1
cudnnDestroy	hipdnnDestroy	
cudnnDestroyActivationDescriptor	hipdnnDestroyActivationDescriptor	
cudnnDestroyAttnDescriptor		7.5.0
cudnnDestroyCTCLossDescriptor		7.0.5
cudnnDestroyConvolutionDescriptor	hipdnnDestroyConvolutionDescriptor	
cudnnDestroyDropoutDescriptor	hipdnnDestroyDropoutDescriptor	
cudnnDestroyFilterDescriptor	hipdnnDestroyFilterDescriptor	
cudnnDestroyFusedOpsConstParamPack		7.6.0
cudnnDestroyFusedOpsPlan		7.6.0
cudnnDestroyFusedOpsVariantParamPack		7.6.0
cudnnDestroyLRNDescriptor	hipdnnDestroyLRNDescriptor	
cudnnDestroyOpTensorDescriptor	hipdnnDestroyOpTensorDescriptor	
cudnnDestroyPoolingDescriptor	hipdnnDestroyPoolingDescri	

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cudnnDestroyRNNDataDescriptor		7.2.1
cudnnDestroyRNNDescriptor	hipdnnDestroyRNNDescriptor	
cudnnDestroyReduceTensorDescriptor	hipdnnDestroyReduceTensorDescriptor	
cudnnDestroySeqDataDescriptor		7.5.0
cudnnDestroySpatialTransformerDescriptor		5.0.0
cudnnDestroyTensorDescriptor	hipdnnDestroyTensorDescriptor	
cudnnDestroyTensorTransformDescriptor		7.5.0
cudnnDivisiveNormalizationBackward		3.0.0
cudnnDivisiveNormalizationForward		3.0.0
cudnnDropoutBackward		5.0.0
cudnnDropoutForward		5.0.0
cudnnDropoutGetReserveSpaceSize		5.0.0
cudnnDropoutGetStatesSize	hipdnnDropoutGetStatesSize	
cudnnFindConvolutionBackwardDataAlgorithm	hipdnnFindConvolutionBackwardDataAlgorithm	
cudnnFindConvolutionBackwardDataAlgorithmEx	hipdnnFindConvolutionBackwardDataAlgorithmEx	
cudnnFindConvolutionBackwardFilterAlgorithm	hipdnnFindConvolutionBackwardFilterAlgorithm	
cudnnFindConvolutionBackwardFilterAlgorithmEx	hipdnnFindConvolutionBackwardFilterAlgorithmEx	

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cudnnFindConvolutionForwardAlgorithm	hipdnnFindConvolutionForwardAlgorithm	
cudnnFindConvolutionForwardAlgorithmEx	hipdnnFindConvolutionForwardAlgorithmEx	
cudnnFusedOpsExecute		7.6.0
cudnnGetActivationDescriptor	hipdnnGetActivationDescriptor	
cudnnGetActivationDescriptorSwishBeta		8.2.0
cudnnGetAttnDescriptor		7.5.0
cudnnGetBatchNormalizationBackwardExWorkspaceSize		7.4.1
cudnnGetBatchNormalizationForwardTrainingExWorkspaceSize		7.4.1
cudnnGetBatchNormalizationTrainingExReserveSpaceSize		7.4.1
cudnnGetCTCLossDescriptor		7.0.5
cudnnGetCTCLossDescriptorEx		7.5.0
cudnnGetCTCLossDescriptor_v8		8.0.1
cudnnGetCTCLossWorkspaceSize		7.0.5
cudnnGetCTCLossWorkspaceSize_v8		8.0.1
cudnnGetCallback		7.1.3
cudnnGetConvolution2dDescriptor	hipdnnGetConvolution2dDescriptor	
cudnnGetConvolution2dForwardOutputDim	hipdnnGetConvolution2dForwardOutputDim	
cudnnGetConvolutionBackwardDataAlgorithmMaxCount		7.0.5

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cudnnGetConvolutionBackwardDataAlgorithm_v7		7.0.5
cudnnGetConvolutionBackwardDataWorkspaceSize	hipdnnGetConvolutionBackwardDataWorkspaceSize	
cudnnGetConvolutionBackwardFilterAlgorithmMaxCount		7.0.5
cudnnGetConvolutionBackwardFilterAlgorithm_v7		7.0.5
cudnnGetConvolutionBackwardFilterWorkspaceSize	hipdnnGetConvolutionBackwardFilterWorkspaceSize	
cudnnGetConvolutionForwardAlgorithmMaxCount		7.0.5
cudnnGetConvolutionForwardAlgorithm_v7		7.0.5
cudnnGetConvolutionForwardWorkspaceSize	hipdnnGetConvolutionForwardWorkspaceSize	
cudnnGetConvolutionGroupCount		7.0.5
cudnnGetConvolutionMathType		7.0.5
cudnnGetConvolutionNdDescriptor		2.0.0
cudnnGetConvolutionNdForwardOutputDim		2.0.0
cudnnGetConvolutionReorderType		7.6.0
cudnnGetCudartVersion		6.0.0
cudnnGetDropoutDescriptor		7.0.5
cudnnGetErrorString	hipdnnGetErrorString	
cudnnGetFilter4dDescriptor	hipdnnGetFilter4dDescriptor	
cudnnGetFilterNdDescriptor	hipdnnGetFilterNdDescriptor	
cudnnGetFilterSizeInBytes		7.6.0

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cudnnGetFoldedConvBackwardDataDescriptors		7.6.0
cudnnGetFusedOpsConstParamPackAttribute		7.6.0
cudnnGetFusedOpsVariantParamPackAttribute		7.6.0
cudnnGetLRNDescriptor	hipdnnGetLRNDescriptor	
cudnnGetMultiHeadAttnBuffers		7.5.0
cudnnGetMultiHeadAttnWeights		7.5.0
cudnnGetNormalizationBackwardWorkspaceSize		8.0.1
cudnnGetNormalizationForwardTrainingWorkspaceSize		8.0.1
cudnnGetNormalizationTrainingReserveSpaceSize		8.0.1
cudnnGetOpTensorDescriptor	hipdnnGetOpTensorDescriptor	
cudnnGetPooling2dDescriptor	hipdnnGetPooling2dDescriptor	
cudnnGetPooling2dForwardOutputDim	hipdnnGetPooling2dForwardOutputDim	
cudnnGetPoolingNdDescriptor		2.0.0
cudnnGetPoolingNdForwardOutputDim		2.0.0
cudnnGetProperty		6.0.0
cudnnGetRNNDataDescriptor		7.2.1
cudnnGetRNNDescriptor_v8		8.0.1
cudnnGetRNNTempSpaceSizes		8.0.1

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cudnnGetRNNTrainingReserveSize	hipdnnGetRNNTrainingReserveSize	
cudnnGetRNNWeightParams		8.0.1
cudnnGetRNNWeightSpaceSize		8.0.1
cudnnGetRNNWorkspaceSize	hipdnnGetRNNWorkspaceSize	
cudnnGetReduceTensorDescriptor	hipdnnGetReduceTensorDescriptor	
cudnnGetReductionIndicesSize		6.0.0
cudnnGetReductionWorkspaceSize	hipdnnGetReductionWorkspaceSize	
cudnnGetSeqDataDescriptor		7.5.0
cudnnGetStream	hipdnnGetStream	
cudnnGetTensor4dDescriptor	hipdnnGetTensor4dDescriptor	
cudnnGetTensorNdDescriptor	hipdnnGetTensorNdDescriptor	
cudnnGetTensorSizeInBytes		6.0.0
cudnnGetTensorTransformDescriptor		7.5.0
cudnnGetVersion	hipdnnGetVersion	
cudnnIm2Col		2.0.0
cudnnInitTransformDest		7.5.0
cudnnLRNCrossChannelBackward	hipdnnLRNCrossChannelBackward	
cudnnLRNCrossChannelForward	hipdnnLRNCrossChannelForward	
cudnnMakeFusedOpsPlan		7.6.0
cudnnMultiHeadAttnBackwardData		7.5.0

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cudnnMultiHeadAttnBackwardWeights		7.5.0
cudnnMultiHeadAttnForward		7.5.0
cudnnNormalizationBackward		8.0.1
cudnnNormalizationForwardInference		8.0.1
cudnnNormalizationForwardTraining		8.0.1
cudnnOpTensor	hipdnnOpTensor	
cudnnOpsInferVersionCheck		8.0.1
cudnnOpsTrainVersionCheck		8.0.1
cudnnPoolingBackward	hipdnnPoolingBackward	
cudnnPoolingForward	hipdnnPoolingForward	
cudnnQueryRuntimeError		7.0.5
cudnnRNNBackwardData	hipdnnRNNBackwardData	
cudnnRNNBackwardData_v8		8.0.2
cudnnRNNBackwardWeights	hipdnnRNNBackwardWeights	
cudnnRNNBackwardWeights_v8		8.0.2
cudnnRNNForward		8.0.1
cudnnRNNForwardInference	hipdnnRNNForwardInference	
cudnnRNNForwardTraining	hipdnnRNNForwardTraining	
cudnnRNNGetClip_v8		8.0.1
cudnnRNNSetClip_v8		8.0.1
cudnnReduceTensor	hipdnnReduceTensor	
cudnnReorderFilterAndBias		7.6.0
cudnnRestoreDropoutDescriptor		7.0.5
cudnnScaleTensor	hipdnnScaleTensor	
cudnnSetActivationDescriptor	hipdnnSetActivationDescriptor	

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cudnnSetActivationDescriptorSwishBeta		8.2.0
cudnnSetAttnDescriptor		7.5.0
cudnnSetCTCLossDescriptor		7.0.5
cudnnSetCTCLossDescriptorEx		7.6.0
cudnnSetCTCLossDescriptor_v8		8.0.1
cudnnSetCallback		7.1.3
cudnnSetConvolution2dDescriptor	hipdnnSetConvolution2dDescriptor	
cudnnSetConvolutionGroupCount	hipdnnSetConvolutionGroupCount	
cudnnSetConvolutionMathType	hipdnnSetConvolutionMathType	
cudnnSetConvolutionNdDescriptor	hipdnnSetConvolutionNdDescriptor	
cudnnSetConvolutionReorderType		7.6.0
cudnnSetDropoutDescriptor	hipdnnSetDropoutDescriptor	
cudnnSetFilter4dDescriptor	hipdnnSetFilter4dDescriptor	
cudnnSetFilterNdDescriptor	hipdnnSetFilterNdDescriptor	
cudnnSetFusedOpsConstParamPackAttribute		7.6.0
cudnnSetFusedOpsVariantParamPackAttribute		7.6.0
cudnnSetLRNDescriptor	hipdnnSetLRNDescriptor	
cudnnSetOpTensorDescriptor	hipdnnSetOpTensorDescriptor	
cudnnSetPooling2dDescriptor	hipdnnSetPooling2dDescriptor	

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cudnnSetPoolingNdDescriptor	hipdnnSetPoolingNdDescriptor	
cudnnSetRNNDataDescriptor		7.2.1
cudnnSetRNNDescriptor_v8		8.0.1
cudnnSetReduceTensorDescriptor	hipdnnSetReduceTensorDescriptor	
cudnnSetSeqDataDescriptor		7.5.0
cudnnSetSpatialTransformerNdDescriptor		5.0.0
cudnnSetStream	hipdnnSetStream	
cudnnSetTensor	hipdnnSetTensor	
cudnnSetTensor4dDescriptor	hipdnnSetTensor4dDescriptor	
cudnnSetTensor4dDescriptorEx	hipdnnSetTensor4dDescriptorEx	
cudnnSetTensorNdDescriptor	hipdnnSetTensorNdDescriptor	
cudnnSetTensorNdDescriptorEx		6.0.0
cudnnSetTensorTransformDescriptor		7.5.0
cudnnSoftmaxBackward	hipdnnSoftmaxBackward	
cudnnSoftmaxForward	hipdnnSoftmaxForward	
cudnnSpatialTfGridGeneratorBackward		5.0.0
cudnnSpatialTfGridGeneratorForward		5.0.0
cudnnSpatialTfSamplerBackward		5.0.0
cudnnSpatialTfSamplerForward		5.0.0
cudnnTransformFilter		7.6.0
cudnnTransformTensor		2.0.0
cudnnTransformTensorEx		7.5.0

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5.4 FFT

5.4.1 Data types

CUDA	HIP	CUDAversion*
CUFFT_ALLOC_FAILED	HIPFFT_ALLOC_FAILED	
CUFFT_C2C	HIPFFT_C2C	
CUFFT_C2R	HIPFFT_C2R	
CUFFT_CB_LD_COMPLEX	HIPFFT_CB_LD_COMPLEX	
CUFFT_CB_LD_COMPLEX_DOUBLE	HIPFFT_CB_LD_COMPLEX_DOUBLE	
CUFFT_CB_LD_REAL	HIPFFT_CB_LD_REAL	
CUFFT_CB_LD_REAL_DOUBLE	HIPFFT_CB_LD_REAL_DOUBLE	
CUFFT_CB_ST_COMPLEX	HIPFFT_CB_ST_COMPLEX	
CUFFT_CB_ST_COMPLEX_DOUBLE	HIPFFT_CB_ST_COMPLEX_DOUBLE	
CUFFT_CB_ST_REAL	HIPFFT_CB_ST_REAL	
CUFFT_CB_ST_REAL_DOUBLE	HIPFFT_CB_ST_REAL_DOUBLE	
CUFFT_CB_UNDEFINED	HIPFFT_CB_UNDEFINED	
CUFFT_COMPATIBILITY_DEFAULT		
CUFFT_COMPATIBILITY_FFTW_PADDING		
CUFFT_COPY_DEVICE_TO_DEVICE		
CUFFT_COPY_DEVICE_TO_HOST		
CUFFT_COPY_HOST_TO_DE		

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VICE		
CUFFT_COPY_UNDEFINED		
CUFFT_D2Z	HIPFFT_D2Z	
CUFFT_EXEC_FAILED	HIPFFT_EXEC_FAILED	
CUFFT_FORMAT_UNDEFINED		
CUFFT_FORWARD	HIPFFT_FORWARD	
CUFFT_INCOMPLETE_PARAMETER_LIST	HIPFFT_INCOMPLETE_PARAMETER_LIST	
CUFFT_INTERNAL_ERROR	HIPFFT_INTERNAL_ERROR	
CUFFT_INVALID_DEVICE	HIPFFT_INVALID_DEVICE	
CUFFT_INVALID_PLAN	HIPFFT_INVALID_PLAN	
CUFFT_INVALID_SIZE	HIPFFT_INVALID_SIZE	
CUFFT_INVALID_TYPE	HIPFFT_INVALID_TYPE	
CUFFT_INVALID_VALUE	HIPFFT_INVALID_VALUE	
CUFFT_INVERSE	HIPFFT_BACKWARD	
CUFFT_LICENSE_ERROR		
CUFFT_NOT_IMPLEMENTED	HIPFFT_NOT_IMPLEMENTED	
CUFFT_NOT_SUPPORTED	HIPFFT_NOT_SUPPORTED	8
CUFFT_NO_WORKSPACE	HIPFFT_NO_WORKSPACE	
CUFFT_PARSE_ERROR	HIPFFT_PARSE_ERROR	
CUFFT_QUERY_1D_FACTORS		
CUFFT_QUERY_UNDEFINED		
CUFFT_R2C	HIPFFT_R2C	

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CUFFT_SETUP_FAILED	HIPFFT_SETUP_FAILED	
CUFFT_SUCCESS	HIPFFT_SUCCESS	
CUFFT_UNALIGNED_DATA	HIPFFT_UNALIGNED_DATA	
CUFFT_VERSION		10.2
CUFFT_VER_BUILD		10.2
CUFFT_VER_MAJOR	HIPFFT_MAJOR_VERSION	10.2
CUFFT_VER_MINOR	HIPFFT_MINOR_VERSION	10.2
CUFFT_VER_PATCH	HIPFFT_PATCH_LEVEL	10.2
CUFFT_WORKAREA_MINIMAL		9.2
CUFFT_WORKAREA_PERFORMANCE		
CUFFT_WORKAREA_USER		9.2
CUFFT_XT_FORMAT_1D_INPUT_SHUFFLED		
CUFFT_XT_FORMAT_INPLACE		
CUFFT_XT_FORMAT_INPLACE_SHUFFLED		
CUFFT_XT_FORMAT_INPUT		
CUFFT_XT_FORMAT_OUTPUT		
CUFFT_Z2D	HIPFFT_Z2D	
CUFFT_Z2Z	HIPFFT_Z2Z	
MAX_CUFFT_ERROR		
cuFFTFORWARD	HIPFFT_FORWARD	10.2
cuFFTINVERSE	HIPFFT_BACKWARD	10.2

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cufftCompatibility		
cufftCompatibility_t		
cufftComplex	hipfftComplex	
cufftDoubleComplex	hipfftDoubleComplex	
cufftDoubleReal	hipfftDoubleReal	
cufftHandle	hipfftHandle	
cufftReal	hipfftReal	
cufftResult	hipfftResult	
cufftResult_t	hipfftResult_t	
cufftType	hipfftType	
cufftType_t	hipfftType_t	
cufftXt1dFactors		
cufftXt1dFactors_t		
cufftXtCallbackType	hipfftXtCallbackType	
cufftXtCallbackType_t	hipfftXtCallbackType_t	
cufftXtCopyType		
cufftXtCopyType_t		
cufftXtQueryType		
cufftXtQueryType_t		
cufftXtSubFormat		
cufftXtSubFormat_t		
cufftXtWorkAreaPolicy		9.2
cufftXtWorkAreaPolicy_t		9.2

5.4.2 APIfunctions

CUDA	HIP	CUDAversion*
cufftCallbackLoadC	hipfftCallbackLoadC	
cufftCallbackLoadD	hipfftCallbackLoadD	

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cufftCallbackLoadR	hipfftCallbackLoadR	
cufftCallbackLoadZ	hipfftCallbackLoadZ	
cufftCallbackStoreC	hipfftCallbackStoreC	
cufftCallbackStoreD	hipfftCallbackStoreD	
cufftCallbackStoreR	hipfftCallbackStoreR	
cufftCallbackStoreZ	hipfftCallbackStoreZ	
cufftCreate	hipfftCreate	
cufftDestroy	hipfftDestroy	
cufftEstimate1d	hipfftEstimate1d	
cufftEstimate2d	hipfftEstimate2d	
cufftEstimate3d	hipfftEstimate3d	
cufftEstimateMany	hipfftEstimateMany	
cufftExecC2C	hipfftExecC2C	
cufftExecC2R	hipfftExecC2R	
cufftExecD2Z	hipfftExecD2Z	
cufftExecR2C	hipfftExecR2C	
cufftExecZ2D	hipfftExecZ2D	
cufftExecZ2Z	hipfftExecZ2Z	
cufftGetProperty	hipfftGetProperty	8
cufftGetSize	hipfftGetSize	
cufftGetSize1d	hipfftGetSize1d	
cufftGetSize2d	hipfftGetSize2d	
cufftGetSize3d	hipfftGetSize3d	
cufftGetSizeMany	hipfftGetSizeMany	
cufftGetSizeMany64	hipfftGetSizeMany64	7.5
cufftGetVersion	hipfftGetVersion	
cufftMakePlan1d	hipfftMakePlan1d	
cufftMakePlan2d	hipfftMakePlan2d	

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cufftMakePlan3d	hipfftMakePlan3d	
cufftMakePlanMany	hipfftMakePlanMany	
cufftMakePlanMany64	hipfftMakePlanMany64	7.5
cufftPlan1d	hipfftPlan1d	
cufftPlan2d	hipfftPlan2d	
cufftPlan3d	hipfftPlan3d	
cufftPlanMany	hipfftPlanMany	
cufftSetAutoAllocation	hipfftSetAutoAllocation	
cufftSetStream	hipfftSetStream	
cufftSetWorkArea	hipfftSetWorkArea	
cufftXtClearCallback	hipfftXtClearCallback	
cufftXtExec		8
cufftXtExecDescriptor		
cufftXtExecDescriptorC2C		
cufftXtExecDescriptorC2R		
cufftXtExecDescriptorD2Z		
cufftXtExecDescriptorR2C		
cufftXtExecDescriptorZ2D		
cufftXtExecDescriptorZ2Z		
cufftXtFree		
cufftXtGetSizeMany		8
cufftXtMakePlanMany		8
cufftXtMalloc		
cufftXtMemcpy		
cufftXtQueryPlan		
cufftXtSetCallback	hipfftXtSetCallback	
cufftXtSetCallbackSharedSize	hipfftXtSetCallbackSharedSize	

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cufftXtSetGPUs		
cufftXtSetWorkArea		
cufftXtSetWorkAreaPolicy		9.2

5.5 SOLVER

CUDA	HIP	CUDA version *
cusolverDnCreate	hipsolverDnCreate	10.2
cusolverDnDestroy	hipsolverDnDestroy	
cusolverDnSetStream	hipsolverDnSetStream	
cusolverDnGetStream	hipsolverDnGetStream	
cusolverDnCreateSyevjInfo	hipsolverDnCreateSyevjInfo	
cusolverDnDestroySyevjInfo	hipsolverDnDestroySyevjInfo	
cusolverDnXsyevjSetTolerance	hipsolverDnXsyevjSetTolerance	
cusolverDnXsyevjSetMaxSweeps	hipsolverDnXsyevjSetMaxSweeps	
cusolverDnXsyevjSetSortEig	hipsolverDnXsyevjSetSortEig	
cusolverDnXsyevjGetResidual	hipsolverDnXsyevjGetResidual	
cusolverDnXsyevjGetSweeps	hipsolverDnXsyevjGetSweeps	
cusolverDnCreateGesvdjInfo	hipsolverDnCreateGesvdjInfo	
cusolverDnDestroyGesvdjInfo	hipsolverDnDestroyGesvdjInfo	
cusolverDnXgesvdjSetTolerance	hipsolverDnXgesvdjSetTolerance	
cusolverDnXgesvdjSetMaxSweeps	hipsolverDnXgesvdjSetMaxSweeps	
cusolverDnXgesvdjSetSortEig	hipsolverDnXgesvdjSetSortEig	
cusolverDnXgesvdjGetResidual	hipsolverDnXgesvdjGetResidual	
cusolverDnXgesvdjGetSweeps	hipsolverDnXgesvdjGetSweeps	
cusolverDnIRSParamsCreate		
cusolverDnIRSParamsDestroy		
cusolverDnIRSParamsSetSolverPre		

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cisions		
cusolverDnIRSParamsSetSolverMainPrecision		
cusolverDnIRSParamsSetSolverLowestPrecision		
cusolverDnIRSParamsSetRefinementSolver		
cusolverDnIRSParamsSetTol		
cusolverDnIRSParamsSetTolInner		
cusolverDnIRSParamsSetMaxIters		
cusolverDnIRSParamsSetMaxItersInner		
cusolverDnIRSParamsEnableFallback		
cusolverDnIRSParamsDisableFallback		
cusolverDnIRSParamsGetMaxIters		
cusolverDnIRSInfosCreate		
cusolverDnIRSInfosDestroy		
cusolverDnIRSInfosGetMaxIters		
cusolverDnIRSInfosGetNiters		
cusolverDnIRSInfosGetOuterNiters		
cusolverDnIRSInfosRequestResidual		
cusolverDnIRSInfosGetResidualHistory		
cusolverDnCreateParams		
cusolverDnDestroyParams		

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cusolverDnSetAdvOptions		
cusolverDnSpotrf_bufferSize	hipsolverDnSpotrf_bufferSize	
cusolverDnDpotrf_bufferSize	hipsolverDnDpotrf_bufferSize	
cusolverDnCpotrf_bufferSize	hipsolverDnCpotrf_bufferSize	
cusolverDnZpotrf_bufferSize	hipsolverDnZpotrf_bufferSize	
cusolverDnSpotrf	hipsolverDnSpotrf	
cusolverDnDpotrf	hipsolverDnDpotrf	
cusolverDnCpotrf	hipsolverDnCpotrf	
cusolverDnZpotrf	hipsolverDnZpotrf	
cusolverDnSpotrs	hipsolverDnSpotrs	
cusolverDnDpotrs	hipsolverDnDpotrs	
cusolverDnCpotrs	hipsolverDnCpotrs	
cusolverDnZpotrs	hipsolverDnZpotrs	
cusolverDnSpotri_bufferSize	hipsolverDnSpotri_bufferSize	
cusolverDnDpotri_bufferSize	hipsolverDnDpotri_bufferSize	
cusolverDnCpotri_bufferSize	hipsolverDnCpotri_bufferSize	
cusolverDnZpotri_bufferSize	hipsolverDnZpotri_bufferSize	
cusolverDnSpotri	hipsolverDnSpotri	
cusolverDnDpotri	hipsolverDnDpotri	
cusolverDnCpotri	hipsolverDnCpotri	
cusolverDnZpotri	hipsolverDnZpotri	
cusolverDnSgetrf_bufferSize	hipsolverDnSgetrf_bufferSize	
cusolverDnDgetrf_bufferSize	hipsolverDnDgetrf_bufferSize	
cusolverDnCgetrf_bufferSize	hipsolverDnCgetrf_bufferSize	
cusolverDnZgetrf_bufferSize	hipsolverDnZgetrf_bufferSize	
cusolverDnSgetrf	hipsolverDnSgetrf	
cusolverDnDgetrf	hipsolverDnDgetrf	
cusolverDnCgetrf	hipsolverDnCgetrf	

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cusolverDnZgetrf	hipsolverDnZgetrf	
cusolverDnSgetrs	hipsolverDnSgetrs	
cusolverDnDgetrs	hipsolverDnDgetrs	
cusolverDnCgetrs	hipsolverDnCgetrs	
cusolverDnZgetrs	hipsolverDnZgetrs	
cusolverDnZZgesv_bufferSize	hipsolverDnZZgesv_bufferSize	
cusolverDnZCgesv_bufferSize		
cusolverDnZKgesv_bufferSize		
cusolverDnZEgesv_bufferSize		
cusolverDnZYgesv_bufferSize		
cusolverDnCCgesv_bufferSize	hipsolverCCgesv_bufferSize	
cusolverDnCKgesv_bufferSize		
cusolverDnCEgesv_bufferSize		
cusolverDnCYgesv_bufferSize		
cusolverDnDDgesv_bufferSize	hipsolverDnDDgesv_bufferSize	
cusolverDnDSgesv_bufferSize		
cusolverDnDHgesv_bufferSize		
cusolverDnDBgesv_bufferSize		
cusolverDnDXgesv_bufferSize		
cusolverDnSSgesv_bufferSize	hipsolverDnSSgesv_bufferSize	
cusolverDnSHgesv_bufferSize		
cusolverDnSBgesv_bufferSize		
cusolverDnSXgesv_bufferSize		
cusolverDnZZgesv	hipsolverDnZZgesv	
cusolverDnZCgesv		
cusolverDnZKgesv		
cusolverDnZEgesv		
cusolverDnZYgesv		

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cusolverDnCCgesv	hipsolverDnCCgesv	
cusolverDnCKgesv		
cusolverDnCEgesv		
cusolverDnCYgesv		
cusolverDnDDgesv	hipsolverDnDDgesv	
cusolverDnDSgesv		
cusolverDnDHgesv		
cusolverDnDBgesv		
cusolverDnDXgesv		
cusolverDnSSgesv	hipsolverDnSSgesv	
cusolverDnSHgesv		
cusolverDnSBgesv		9
cusolverDnSXgesv		
cusolverDnIRSXgesv_bufferSize		
cusolverDnIRSXgesv		
cusolverDnSgeqrf_bufferSize	hipsolverDnSgeqrf_bufferSize	
cusolverDnDgeqrf_bufferSize	hipsolverDnDgeqrf_bufferSize	
cusolverDnCgeqrf_bufferSize	hipsolverDnCgeqrf_bufferSize	
cusolverDnZgeqrf_bufferSize	hipsolverDnZgeqrf_bufferSize	
cusolverDnSgeqrf	hipsolverDnSgeqrf	
cusolverDnDgeqrf	hipsolverDnDgeqrf	
cusolverDnCgeqrf	hipsolverDnCgeqrf	
cusolverDnZgeqrf	hipsolverDnZgeqrf	
cusolverDnZZgels_bufferSize	hipsolverDnZZgels_bufferSize	
cusolverDnZCgels_bufferSize		
cusolverDnZKgels_bufferSize		
cusolverDnZEgels_bufferSize		
cusolverDnZYgels_bufferSize		

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cusolverDnCCgels_bufferSize	hipsolverDnCCgels_bufferSize	
cusolverDnCKgels_bufferSize		
cusolverDnCEgels_bufferSize		
cusolverDnCYgels_bufferSize		
cusolverDnDDgels_bufferSize	hipsolverDnDDgels_bufferSize	
cusolverDnDSgels_bufferSize		
cusolverDnDHgels_bufferSize		
cusolverDnDBgels_bufferSize		
cusolverDnDXgels_bufferSize		
cusolverDnSSgels_bufferSize	hipsolverDnSSgels_bufferSize	
cusolverDnSHgels_bufferSize		
cusolverDnSBgels_bufferSize		
cusolverDnSXgels_bufferSize		
cusolverDnZZgels	hipsolverDnZZgels	
cusolverDnZCgels		
cusolverDnZKgels		
cusolverDnZEgels		
cusolverDnZYgels		
cusolverDnCCgels	hipsolverDnCCgels	
cusolverDnCKgels		
cusolverDnCEgels		
cusolverDnCYgels		
cusolverDnDDgels	hipsolverDnDDgels	
cusolverDnDSgels		
cusolverDnDHgels		
cusolverDnDBgels		
cusolverDnDXgels		
cusolverDnSSgels	hipsolverDnSSgels	

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cusolverDnSHgels		
cusolverDnSBgels		
cusolverDnSXgels		
cusolverDnIRSXgels_bufferSize		
cusolverDnIRSXgels		
cusolverDnSormqr_bufferSize	hipsolverDnSormqr_bufferSize	
cusolverDnDormqr_bufferSize	hipsolverDnDormqr_bufferSize	
cusolverDnCunmqr_bufferSize	hipsolverDnCunmqr_bufferSize	
cusolverDnZunmqr_bufferSize	hipsolverDnZunmqr_bufferSize	
cusolverDnSormqr	hipsolverDnSormqr	
cusolverDnDormqr	hipsolverDnDormqr	
cusolverDnCunmqr	hipsolverDnCunmqr	
cusolverDnZunmqr	hipsolverDnZunmqr	
cusolverDnSorgqr_bufferSize	hipsolverDnSorgqr_bufferSiz	
cusolverDnDorgqr_bufferSize	hipsolverDnDorgqr_bufferSiz	
cusolverDnCungqr_bufferSize	hipsolverDnCungqr_bufferSize	
cusolverDnZungqr_bufferSize	hipsolverDnZungqr_bufferSize	
cusolverDnSorgqr	hipsolverDnSorgqr	
cusolverDnDorgqr	hipsolverDnDorgqr	
cusolverDnCungqr	hipsolverDnCungqr	
cusolverDnZungqr	hipsolverDnZungqr	
cusolverDnSsytrf_bufferSize	hipsolverDnSsytrf_bufferSiz	
cusolverDnDsytrf_bufferSize	hipsolverDnDsytrf_bufferSiz	
cusolverDnCsytrf_bufferSize	hipsolverDnCsytrf_bufferSiz	
cusolverDnZsytrf_bufferSize	hipsolverDnZsytrf_bufferSiz	
cusolverDnSsytrf	hipsolverDnSsytrf	
cusolverDnDsytrf	hipsolverDnDsytrf	
cusolverDnCsytrf	hipsolverDnCsytrf	

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cusolverDnZsytrf	hipsolverDnZsytrf	
cusolverDnSpotrfBatched	hipsolverDnSpotrfBatched	
cusolverDnDpotrfBatched	hipsolverDnDpotrfBatched	
cusolverDnCpotrfBatched	hipsolverDnCpotrfBatched	
cusolverDnZpotrfBatched	hipsolverDnZpotrfBatched	
cusolverDnSpotrsBatched	hipsolverDnSpotrsBatched	
cusolverDnDpotrsBatched	hipsolverDnDpotrsBatched	
cusolverDnCpotrsBatched	hipsolverDnCpotrsBatched	
cusolverDnZpotrsBatched	hipsolverDnZpotrsBatched	
cusolverDnSgebrd_bufferSize	hipsolverDnSgebrd_bufferSize	
cusolverDnDgebrd_bufferSize	hipsolverDnDgebrd_bufferSize	
cusolverDnCgebrd_bufferSize	hipsolverDnCgebrd_bufferSize	
cusolverDnZgebrd_bufferSize	hipsolverDnZgebrd_bufferSize	
cusolverDnSgebrd	hipsolverDnSgebrd	
cusolverDnDgebr	hipsolverDnDgebrd	
cusolverDnCgebrd	hipsolverDnCgebrd	
cusolverDnZgebrd	hipsolverDnZgebrd	
cusolverDnSorgbr_bufferSize	hipsolverDnSorgbr_bufferSize	
cusolverDnDorgbr_bufferSize	hipsolverDnDorgbr_bufferSize	
cusolverDnCungbr_bufferSize	hipsolverDnCungbr_bufferSize	
cusolverDnZungbr_bufferSize	hipsolverDnZungbr_bufferSize	
cusolverDnSorgbr	hipsolverDnSorgbr	
cusolverDnDorgbr	hipsolverDnDorgbr	
cusolverDnCungbr	hipsolverDnCungbr	
cusolverDnZungbr	hipsolverDnZungbr	
cusolverDnSsytrd_bufferSize	hipsolverDnSsytrd_bufferSize	
cusolverDnDsytrd_bufferSize	hipsolverDnDsytrd_bufferSize	
cusolverDnChetrd_bufferSize	hipsolverDnChetrd_bufferSize	

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cusolverDnZhetrd_bufferSize	hipsolverDnZhetrd_bufferSize	
cusolverDnSsytrd	hipsolverDnSsytrd	
cusolverDnDsytrd	hipsolverDnDsytrd	
cusolverDnChetrd	hipsolverDnChetrd	
cusolverDnZhetrd	hipsolverDnZhetrd	
cusolverDnSormtr_bufferSize	hipsolverDnSormtr_bufferSize	
cusolverDnDormtr_bufferSize	hipsolverDnDormtr_bufferSize	
cusolverDnCunmtr_bufferSize	hipsolverDnCunmtr_bufferSize	
cusolverDnZunmtr_bufferSize	hipsolverDnZunmtr_bufferSize	
cusolverDnSormtr	hipsolverDnSormtr	
cusolverDnDormtr	hipsolverDnDormtr	
cusolverDnCunmtr	hipsolverDnCunmtr	
cusolverDnZunmtr	hipsolverDnZunmtr	
cusolverDnSorgtr_bufferSiz	hipsolverDnSorgtr_bufferSize	
cusolverDnDorgtr_bufferSize	hipsolverDnDorgtr_bufferSize	
cusolverDnCungtr_bufferSize	hipsolverDnCungtr_bufferSize	
cusolverDnZungtr_bufferSize	hipsolverDnZungtr_bufferSize	
cusolverDnSorgtr	hipsolverDnSorgtr	
cusolverDnDorgtr	hipsolverDnDorgtr	
cusolverDnCungtr	hipsolverDnCungtr	
cusolverDnZungtr	hipsolverDnZungtr	
cusolverDnSgesvd_bufferSize	hipsolverDnSgesvd_bufferSize	
cusolverDnDgesvd_bufferSize	hipsolverDnDgesvd_bufferSize	
cusolverDnCgesvd_bufferSize	hipsolverDnCgesvd_bufferSize	
cusolverDnZgesvd_bufferSize	hipsolverDnZgesvd_bufferSize	
cusolverDnSgesvd	hipsolverDnSgesvd	
cusolverDnDgesvd	hipsolverDnDgesvd	
cusolverDnCgesvd	hipsolverDnCgesvd	

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cusolverDnZgesvd	hipsolverDnZgesvd	
cusolverDnSgesvdj_bufferSize	hipsolverDnSgesvdj_bufferSize	
cusolverDnDgesvdj_bufferSize	hipsolverDnDgesvdj_bufferSize	
cusolverDnCgesvdj_bufferSize	hipsolverDnCgesvdj_bufferSize	
cusolverDnZgesvdj_bufferSize	hipsolverDnZgesvdj_bufferSize	
cusolverDnSgesvdj	hipsolverDnSgesvdj	
cusolverDnDgesvdj	hipsolverDnDgesvdj	
cusolverDnCgesvdj	hipsolverDnCgesvdj	
cusolverDnZgesvdj	hipsolverDnZgesvdj	
cusolverDnSgesvdjBatched_buffer Size	hipsolverDnSgesvdjBatched_buffer Size	
cusolverDnDgesvdjBatched_buffer Size	hipsolverDnDgesvdjBatched_buffe rSize	
cusolverDnCgesvdjBatched_buffer Size	hipsolverDnCgesvdjBatched_buffer Size	
cusolverDnZgesvdjBatched_buffer Size	hipsolverDnZgesvdjBatched_buffer Size	
cusolverDnSgesvdjBatched	hipsolverDnSgesvdjBatched	
cusolverDnDgesvdjBatched	hipsolverDnDgesvdjBatched	
cusolverDnCgesvdjBatched	hipsolverDnCgesvdjBatched	
cusolverDnZgesvdjBatched	hipsolverDnZgesvdjBatched	
cusolverDnSgesvdaStridedBatched _bufferSize	hipsolverDnSgesvdaStridedBatched _bufferSize	
cusolverDnDgesvdaStridedBatched _bufferSize	hipsolverDnDgesvdaStridedBatche d_bufferSize	
cusolverDnCgesvdaStridedBatched _bufferSize	hipsolverDnCgesvdaStridedBatche d_bufferSize	
cusolverDnZgesvdaStridedBatched	hipsolverDnZgesvdaStridedBatche	

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_bufferSize	d_bufferSize	
cusolverDnSgesvdaStridedBatched	hipsolverDnSgesvdaStridedBatched	
cusolverDnDgesvdaStridedBatched	hipsolverDnDgesvdaStridedBatched	
cusolverDnCgesvdaStridedBatched	hipsolverDnCgesvdaStridedBatched	
cusolverDnZgesvdaStridedBatched	hipsolverDnZgesvdaStridedBatched	
cusolverDnSsyevd_bufferSize	hipsolverDnSsyevd_bufferSize	
cusolverDnDsyevd_bufferSize	hipsolverDnDsyevd_bufferSize	
cusolverDnCheevd_bufferSize	hipsolverDnCheevd_bufferSize	
cusolverDnZheevd_bufferSize	hipsolverDnZheevd_bufferSize	
cusolverDnSsyevd	hipsolverDnSsyevd	
cusolverDnDsyevd	hipsolverDnDsyevd	
cusolverDnCheevd	hipsolverDnCheevd	
cusolverDnZheevd	hipsolverDnZheevd	
cusolverDnSsyevdx_bufferSize	hipsolverDnSsyevdx_bufferSize	
cusolverDnDsyevdx_bufferSize	hipsolverDnDsyevdx_bufferSize	
cusolverDnCheevdx_bufferSize	hipsolverDnCheevdx_bufferSize	
cusolverDnZheevdx_bufferSize	hipsolverDnZheevdx_bufferSize	
cusolverDnSsyevdx	hipsolverDnSsyevdx	
cusolverDnDsyevdx	hipsolverDnDsyevdx	
cusolverDnCheevdx	hipsolverDnCheevdx	
cusolverDnZheevdx	hipsolverDnZheevdx	
cusolverDnSsygvd_bufferSize	hipsolverDnSsygvd_bufferSize	
cusolverDnDsygvd_bufferSize	hipsolverDnDsygvd_bufferSize	
cusolverDnChegvd_bufferSize	hipsolverDnChegvd_bufferSize	
cusolverDnZhegvd_bufferSize	hipsolverDnZhegvd_bufferSize	

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cusolverDnSsygvd	hipsolverDnSsygvd	
cusolverDnDsygvd	hipsolverDnDsygvd	
cusolverDnChegvd	hipsolverDnChegvd	
cusolverDnZhegvd	hipsolverDnZhegvd	
cusolverDnSsygvdx_bufferSize	hipsolverDnSsygvdx_bufferSize	
cusolverDnDsygvdx_bufferSize	hipsolverDnDsygvdx_bufferSize	
cusolverDnChegvdx_bufferSize	hipsolverDnChegvdx_bufferSize	
cusolverDnZhegvdx_bufferSize	hipsolverDnZhegvdx_bufferSize	
cusolverDnSsygvdx	hipsolverDnSsygvdx	
cusolverDnDsygvdx	hipsolverDnDsygvdx	
cusolverDnChegvdx	hipsolverDnChegvdx	
cusolverDnZhegvdx	hipsolverDnZhegvdx	
cusolverDnSsyevj_bufferSize	hipsolverDnSsyevj_bufferSize	
cusolverDnDsyevj_bufferSize	hipsolverDnDsyevj_bufferSize	
cusolverDnCheevj_bufferSize	hipsolverDnCheevj_bufferSize	
cusolverDnZheevj_bufferSize	hipsolverDnZheevj_bufferSize	
cusolverDnSsyevj	hipsolverDnSsyevj	
cusolverDnDsyevj	hipsolverDnDsyevj	
cusolverDnCheevj	hipsolverDnCheevj	
cusolverDnZheevj	hipsolverDnZheevj	
cusolverDnSsygvj_bufferSize	hipsolverDnSsygvj_bufferSize	
cusolverDnDsygvj_bufferSize	hipsolverDnDsygvj_bufferSize	
cusolverDnChegvj_bufferSize	hipsolverDnChegvj_bufferSize	
cusolverDnZhegvj_bufferSize	hipsolverDnZhegvj_bufferSize	
cusolverDnSsygvj	hipsolverDnSsygvj	
cusolverDnDsygvj	hipsolverDnDsygvj	
cusolverDnChegvj	hipsolverDnChegvj_	
cusolverDnZhegvj	hipsolverDnZhegvj_	

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cusolverDnSsyevjBatched_bufferSize	hipsolverDnSsyevjBatched_bufferSize	
cusolverDnDsyevjBatched_bufferSize	hipsolverDnDsyevjBatched_bufferSize	
cusolverDnCheevjBatched_bufferSize	hipsolverDnCheevjBatched_bufferSize	
cusolverDnZheevjBatched_bufferSize	hipsolverDnZheevjBatched_bufferSize	
cusolverDnSsyevjBatched	hipsolverDnSsyevjBatched	
cusolverDnDsyevjBatched	hipsolverDnDsyevjBatched	
cusolverDnCheevjBatched	hipsolverDnCheevjBatched	
cusolverDnZheevjBatched	hipsolverDnZheevjBatched	
cusolverDnXpotrf_bufferSize		
cusolverDnXpotrf		
cusolverDnXpotrs		
cusolverDnXgetrf_bufferSize		
cusolverDnXgetrf		
cusolverDnXgetrs		
cusolverDnXgeqrf_bufferSize		
cusolverDnXgeqrf		
cusolverDnXsytrs_bufferSize		
cusolverDnXsytrs		
cusolverDnXtrtri_bufferSize		
cusolverDnXtrtri		
cusolverDnXgesvd_bufferSize		
cusolverDnXgesvd		
cusolverDnXgesvdp_bufferSize		
cusolverDnXgesvdp		

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cusolverDnXgesvdr_bufferSize		
cusolverDnXgesvdr		
cusolverDnXsyevd_bufferSize		
cusolverDnXsyevd		
cusolverDnXsyevdx_bufferSize		
cusolverDnXsyevdx		
cusolverSpCreate		
cusolverSpDestroy		
cusolverSpSetStream		
cusolverSpXcsrissymHost		
cusolverSpScsrlsvlu		
cusolverSpDcsrlsvlu		
cusolverSpCcsrlsvlu		
cusolverSpZcsrlsvlu		
cusolverSpScsrlsvqr		
cusolverSpDcsrlsvqr		
cusolverSpCcsrlsvqr		
cusolverSpZcsrlsvqr		
cusolverSpScsrlsvchol		
cusolverSpDcsrlsvchol		
cusolverSpCcsrlsvchol		
cusolverSpZcsrlsvchol		
cusolverSpScsrlsqvqr		
cusolverSpDcsrlsqvqr		
cusolverSpCcsrlsqvqr		
cusolverSpZcsrlsqvqr		
cusolverSpScsreigvsi		
cusolverSpDcsreigvsi		

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cusolverSpCcsreigvsi		
cusolverSpZcsreigvsi		
solverspScsreigs		
cusolverSpDcsreigs		
cusolverSpCcsreigs		
cusolverSpZcsreigs		
cusolverSpXcsrsymrcmHost		
cusolverSpXcsrsymmdqHost		
cusolverSpXcsrsymamdHost		
cusolverSpXcsrmetisndHost		
cusolverSpScsrzfdHost		
cusolverSpDcsrzfdHost		
cusolverSpCcsrzfdHost		
cusolverSpZcsrzfdHost		
cusolverSpXcsrperm_bufferSizeHost		
cusolverSpXcsrpermHost		
cusolverSpCreateCsrqrInfo		
cusolverSpDestroyCsrqrInfo		
cusolverSpXcsrqrAnalysisBatched		
cusolverSpScsrqrBufferInfoBatched		
cusolverSpDcsrqrBufferInfoBatched		
cusolverSpCcsrqrBufferInfoBatched		
cusolverSpZcsrqrBufferInfoBatched		

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cusolverSpScsrqrsvBatched		
cusolverSpDcsrqrsvBatched		
cusolverSpCcsrqrsvBatched		
cusolverSpZcsrqrsvBatched		
cusolverRfAccessBundledFactors		
cusolverRfAnalyze		
cusolverRfSetupDevice		
cusolverRfSetupHost		
cusolverRfCreate		
cusolverRfExtractBundledFactorsHost		
cusolverRfExtractSplitFactorsHost		
cusolverRfDestroy		
cusolverRfGetMatrixFormat		
cusolverRfGetNumericProperties		
cusolverRfGetNumericBoostReport		
cusolverRfGetResetValuesFastMode		
cusolverRfGet_Algs		
cusolverRfRefactor		
cusolverRfResetValues		
cusolverRfSetMatrixFormat		
cusolverRfSetNumericProperties		
cusolverRfSetResetValuesFastMode		
cusolverRfSetAlgs		
cusolverRfSolve		
cusolverRfBatchSetupHost		

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cusolverRfBatchAnalyze		
cusolverRfBatchResetValues		
cusolverRfBatchRefactor		
cusolverRfBatchSolve		
cusolverRfBatchZeroPivot		

5.6 SPARSE

5.6.1 hipSPARSE 数据类型支持

type	CUDA	HIP	CUDAversion*
define	CUSPARSE_VER_MAJOR		10.1Update2
define	CUSPARSE_VER_MINOR		10.1Update2
define	CUSPARSE_VER_PATCH		10.1Update2
define	CUSPARSE_VER_BUILD		10.1Update2
define	CUSPARSE_VERSION		10.1Update2
enum	cusparseAction_t	hipsparseAction_t	
0	CUSPARSE_ACTION_SYMBOLIC	HIPSPARSE_ACTION_SYMBOLIC	
1	CUSPARSE_ACTION_NUMERIC	HIPSPARSE_ACTION_NUMERIC	
enum	cusparseDirection_t	hipsparseDirection_t	
0	CUSPARSE_DIRECTION_ROW	HIPSPARSE_DIRECTION_ROW	
1	CUSPARSE_DIRECTION_COLUMN	HIPSPARSE_DIRECTION_COLUMN	
enum	cusparseHybPartition_t	hipsparseHybPartition_t	

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0	CUSPARSE_HYB_PARTITION_AUTO	HIPSPARSE_HYB_PARTITION_AUTO	
1	CUSPARSE_HYB_PARTITION_USER	HIPSPARSE_HYB_PARTITION_USER	
2	CUSPARSE_HYB_PARTITION_MAX	HIPSPARSE_HYB_PARTITION_MAX	
enum	cusparseDiagType_t	hipsparseDiagType_t	
0	CUSPARSE_DIAG_TYPE_NON_UNIT	HIPSPARSE_DIAG_TYPE_NON_UNIT	
1	CUSPARSE_DIAG_TYPE_UNIT	HIPSPARSE_DIAG_TYPE_UNIT	
enum	cusparseFillMode_t	hipsparseFillMode_t	
0	CUSPARSE_FILL_MODE_LOWER	HIPSPARSE_FILL_MODE_LOWER	
1	CUSPARSE_FILL_MODE_UPPER	HIPSPARSE_FILL_MODE_UPPER	
enum	cusparseIndexBase_t	hipsparseIndexBase_t	
0	CUSPARSE_INDEX_BASE_ZERO	HIPSPARSE_INDEX_BASE_ZERO	
1	CUSPARSE_INDEX_BASE_ONE	HIPSPARSE_INDEX_BASE_ONE	
enum	cusparseMatrixType_t	hipsparseMatrixType_t	
0	CUSPARSE_MATRIX_TYPE_GENERAL	HIPSPARSE_MATRIX_TYPE_GENERAL	
1	CUSPARSE_MATRIX_TYPE_SYMMETRIC	HIPSPARSE_MATRIX_TYPE_SYMMETRIC	
2	CUSPARSE_MATRIX_TYPE_HERMITIAN	HIPSPARSE_MATRIX_TYPE_HERMITIAN	

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3	CUSPARSE_MATRIX_TYPE_TRIANGULAR	HIPSPARSE_MATRIX_TYPE_TRIANGULAR	
enum	cusparseOperation_t	hipsparseOperation_t	
0	CUSPARSE_OPERATION_NON_TRANSPOSE	HIPSPARSE_OPERATION_NON_TRANSPOSE	
1	CUSPARSE_OPERATION_TRANSPOSE	HIPSPARSE_OPERATION_TRANSPOSE	
2	CUSPARSE_OPERATION_CONJUGATE_TRANSPOSE	HIPSPARSE_OPERATION_CONJUGATE_TRANSPOSE	
enum	cusparsePointerMode_t	hipsparsePointerMode_t	
0	CUSPARSE_POINTER_MODE_HOST	HIPSPARSE_POINTER_MODE_HOST	
1	CUSPARSE_POINTER_MODE_DEVICE	HIPSPARSE_POINTER_MODE_DEVICE	
enum	cusparseAlgMode_t		8.0
0	CUSPARSE_ALG0		8.0
1	CUSPARSE_ALG1		8.0
0	CUSPARSE_ALG_NAIVE		9.2
1	CUSPARSE_ALG_MERGE_PATH		9.2
enum	cusparseSolvePolicy_t	hipsparseSolvePolicy_t	
0	CUSPARSE_SOLVE_POLICY_NO_LEVEL	HIPSPARSE_SOLVE_POLICY_NO_LEVEL	
1	CUSPARSE_SOLVE_POLICY_USE_LEVEL	HIPSPARSE_SOLVE_POLICY_USE_LEVEL	
enum	cusparseStatus_t	hipsparseMatrixType_t	

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0	CUSPARSE_STATUS_SUCCESS	HIPSPARSE_STATUS_SUCCESS	
1	CUSPARSE_STATUS_NOT_INITIALIZED	HIPSPARSE_STATUS_NOT_INITIALIZED	
2	CUSPARSE_STATUS_ALLOC_FAILED	HIPSPARSE_STATUS_ALLOC_FAILED	
3	CUSPARSE_STATUS_INVALID_VALUE	HIPSPARSE_STATUS_INVALID_VALUE	
4	CUSPARSE_STATUS_ARCH_MISMATCH	HIPSPARSE_STATUS_ARCH_MISMATCH	
5	CUSPARSE_STATUS_MAPPING_ERROR	HIPSPARSE_STATUS_MAPPING_ERROR	
6	CUSPARSE_STATUS_EXECUTION_FAILED	HIPSPARSE_STATUS_EXECUTION_FAILED	
7	CUSPARSE_STATUS_INTERNAL_ERROR	HIPSPARSE_STATUS_INTERNAL_ERROR	
8	CUSPARSE_STATUS_MATRIX_TYPE_NOT_SUPPORTED	HIPSPARSE_STATUS_MATRIX_TYPE_NOT_SUPPORTED	
9	CUSPARSE_STATUS_ZERO_PIVOT	HIPSPARSE_STATUS_ZERO_PIVOT	
struct	cusparseContext		
typedef	cusparseHandle_t	hipsparseHandle_t	
struct	cusparseHybMat		
typedef	cusparseHybMat_t	hipsparseHybMat_t	
struct	cusparseMatDescr		
typed	cusparseMatDescr_t	hipsparseMatDescr_t	

ef			
struct	cusparseSolveAnalysisInfo		
typedef	cusparseSolveAnalysisInfo_t		
struct	csrsv2Info		
typedef	csrsv2Info_t	csrsv2Info_t	
struct	csrsm2Info	csrsm2Info	9.2
typedef	csrsm2Info_t	csrsm2Info_t	
struct	bsrsv2Info		
typedef	bsrsv2Info_t		
struct	bsrsm2Info		
typedef	bsrsm2Info_t		
struct	bsric02Info		
typedef	bsric02Info_t		
struct	csrilu02Info		
typedef	csrilu02Info_t	csrilu02Info_t	
struct	bsrilu02Info		
typedef	bsrilu02Info_t		
struct	csru2csrInfo		
typed	csru2csrInfo_t		

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ef			
struct	csrgemm2Info	csrgemm2Info	
typed ef	csrgemm2Info_t	csrgemm2Info_t	
struct	cusparseColorInfo		
typed ef	cusparseColorInfo_t		
struct	pruneInfo		9.0
typed ef	pruneInfo_t		9.0
enum	cusparseCsr2CscAlg_t		10.1
1	CUSPARSE_CSR2CSC_ALG1		10.1
2	CUSPARSE_CSR2CSC_ALG2		10.1
enum	cusparseFormat_t		10.1
1	CUSPARSE_FORMAT_CSR		10.1
2	CUSPARSE_FORMAT_CSC		10.1
3	CUSPARSE_FORMAT_COO		10.1
4	CUSPARSE_FORMAT_COO_AOS		10.1
enum	cusparseOrder_t		10.1
1	CUSPARSE_ORDER_COL		10.1
2	CUSPARSE_ORDER_R		10.1

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	OW		
enum	cusparseSpMVAlg_t		10.1
0	CUSPARSE_MV_ALG_DEFAULT		10.1
1	CUSPARSE_COOMV_ALG		10.1
2	CUSPARSE_CSRMV_ALG1		10.1
3	CUSPARSE_CSRMV_ALG2		10.1
enum	cusparseSpMMAlg_t		10.1
0	CUSPARSE_MM_ALG_DEFAULT		10.1
1	CUSPARSE_COOMM_ALG1		10.1
2	CUSPARSE_COOMM_ALG2		10.1
3	CUSPARSE_COOMM_ALG3		10.1
4	CUSPARSE_CSRMM_ALG1		10.1
enum	cusparseIndexType_t		10.1
1	CUSPARSE_INDEX_16U		10.1
2	CUSPARSE_INDEX_32I		10.1
3	CUSPARSE_INDEX_64I		10.1

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struct	cusparseSpMatDescr		10.1
typedef	cusparseSpMatDescr_t		10.1
struct	cusparseDnMatDescr		10.1
typedef	cusparseDnMatDescr_t		10.1
struct	cusparseSpVecDescr		10.1
typedef	cusparseSpVecDescr_t		10.1
struct	cusparseDnVecDescr		10.1
typedef	cusparseDnVecDescr_t		10.1

5.6.2 hipSPARSE 辅助函数

CUDA	HIP	CUDAversion *
cusparseCreate	hipsparseCreate	
cusparseCreateSolveAnalysisInfo		
cusparseCreateHybMat	hipsparseCreateHybMat	
cusparseCreateMatDescr	hipsparseCreateMatDescr	
cusparseDestroy	hipsparseDestroy	
cusparseDestroySolveAnalysisInfo		
cusparseDestroyHybMat	hipsparseDestroyHybMat	
cusparseDestroyMatDescr	hipsparseDestroyMatDescr	
cusparseGetLevelInfo		
cusparseGetMatDiagType	hipsparseGetMatDiagType	
cusparseGetMatFillMode	hipsparseGetMatFillMode	

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cusparseGetMatIndexBase	hipsparseGetMatIndexBase	
cusparseGetMatType	hipsparseGetMatType	
cusparseGetPointerMode	hipsparseGetPointerMode	
cusparseGetVersion	hipsparseGetVersion	
cusparseSetMatDiagType	hipsparseSetMatDiagType	
cusparseSetMatIndexBase	hipsparseSetMatIndexBase	
cusparseSetMatFillMode	hipsparseSetMatFillMode	
cusparseSetMatType	hipsparseSetMatType	
cusparseSetPointerMode	hipsparseSetPointerMode	
cusparseSetStream	hipsparseSetStream	
cusparseGetStream	hipsparseGetStream	8
cusparseCreateCsrsv2Info	hipsparseCreateCsrsv2Info	
cusparseDestroyCsrsv2Info	hipsparseDestroyCsrsv2Info	
cusparseCreateCsrsm2Info	hipsparseCreateCsrsm2Info	9.2
cusparseDestroyCsrsm2Info	hipsparseDestroyCsrsm2Info	9.2
cusparseCreateCsric02Info	hipsparseCreateCsric02Info	
cusparseDestroyCsric02Info	hipsparseDestroyCsric02Info	
cusparseCreateCsrilu02Info	hipsparseCreateCsrilu02Info	
cusparseDestroyCsrilu02Info	hipsparseDestroyCsrilu02Info	
cusparseCreateBsrsv2Info	hipsparseCreateBsrsv2Info	
cusparseDestroyBsrsv2Info	hipsparseDestroyBsrsv2Info	
cusparseCreateBsrsm2Info	hipsparseCreateBsrsm2Info	
cusparseDestroyBsrsm2Info	hipsparseDestroyBsrsm2Info	
cusparseCreateBsric02Info	hipsparseCreateBsric02Info	
cusparseDestroyBsric02Info	hipsparseDestroyBsric02Info	
cusparseCreateBsrilu02Info	hipsparseCreateBsrilu02Info	
cusparseDestroyBsrilu02Info	hipsparseDestroyBsrilu02Info	
cusparseCreateCsrgemm2Info	hipsparseCreateCsrgemm2Info	

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cusparseDestroyCsrgemm2Info	hipsparseDestroyCsrgemm2Info	
cusparseCreatePruneInfo	hipsparseCreatePruneInfo	9

5.6.3 Level1 支持函数列表

CUDA	HIP	CUDAversion
cusparseSaxpyi	hipsparseSaxpyi	
cusparseDaxpyi	hipsparseDaxpyi	
cusparseCaxpyi	hipsparseCaxpyi	
cusparseZaxpyi	hipsparseZaxpyi	
cusparseSdoti	hipsparseSdoti	
cusparseDdoti	hipsparseDdoti	
cusparseCdoti	hipsparseCdoti	
cusparseZdoti	hipsparseZdoti	
cusparseCdotci	hipsparseCdotci	
cusparseZdotci	hipsparseZdotci	
cusparseSgthr	hipsparseSgthr	
cusparseDgthr	hipsparseDgthr	
cusparseCgthr	hipsparseCgthr	
cusparseZgthr	hipsparseZgthr	
cusparseSgthrz	hipsparseSgthrz	
cusparseDgthrz	hipsparseDgthrz	
cusparseCgthrz	hipsparseCgthrz	
cusparseZgthrz	hipsparseZgthrz	
cusparseSroti	hipsparseSroti	
cusparseDroti	hipsparseDroti	
cusparseSsctr	hipsparseSsctr	
cusparseDsctr	hipsparseDsctr	
cusparseCsctr	hipsparseCsctr	

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cusparseZsctr	hipsparseZsctr	

5.6.4 Level2 支持函数列表

CUDA	HIP	CUDA version*
cusparseSbsrmv	hipsparseSbsrmv	
cusparseDbsrmv	hipsparseDbsrmv	
cusparseCbsrmv	hipsparseCbsrmv	
cusparseZbsrmv	hipsparseZbsrmv	
cusparseSbsrxmv	hipsparseSbsrxmv	
cusparseDbsrxmv	hipsparseDbsrxmv	
cusparseCbsrxmv	hipsparseCbsrxmv	
cusparseZbsrxmv	hipsparseZbsrxmv	
cusparseScsrmv	hipsparseScsrmv	
cusparseDcsrmv	hipsparseDcsrmv	
cusparseCcsrmv	hipsparseCcsrmv	
cusparseZcsrmv	hipsparseZcsrmv	
cusparseCsrmvEx		8
cusparseCsrmvEx_bufferSize		8
cusparseScsrmv_mp		8
cusparseDcsrmv_mp		8
cusparseCcsrmv_mp		8
cusparseZcsrmv_mp		8
cusparseSgemvi	hipsparseSgemvi	7.5
cusparseDgemvi	hipsparseDgemvi	7.5
cusparseCgemvi	hipsparseCgemvi	7.5
cusparseZgemvi	hipsparseZgemvi	7.5
cusparseSgemvi_bufferSize	hipsparseSgemvi_bufferSize	7.5
cusparseDgemvi_bufferSize	hipsparseDgemvi_bufferSize	7.5

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cusparseCgemvi_bufferSize	hipsparseCgemvi_bufferSize	7.5
cusparseZgemvi_bufferSize	hipsparseZgemvi_bufferSize	7.5
cusparseSbsrsv2_bufferSize	hipsparseSbsrsv2_bufferSize	
cusparseSbsrsv2_bufferSizeExt	hipsparseSbsrsv2_bufferSizeExt	
cusparseDbsrsv2_bufferSize	hipsparseDbsrsv2_bufferSize	
cusparseDbsrsv2_bufferSizeExt	hipsparseDbsrsv2_bufferSizeExt	
cusparseCbsrsv2_bufferSize	hipsparseCbsrsv2_bufferSize	
cusparseCbsrsv2_bufferSizeExt	hipsparseCbsrsv2_bufferSizeExt	
cusparseZbsrsv2_bufferSize	hipsparseZbsrsv2_bufferSize	
cusparseZbsrsv2_bufferSizeExt	hipsparseZbsrsv2_bufferSizeExt	
cusparseSbsrsv2_analysis	hipsparseSbsrsv2_analysis	
cusparseDbsrsv2_analysis	hipsparseDbsrsv2_analysis	
cusparseCbsrsv2_analysis	hipsparseCbsrsv2_analysis	
cusparseZbsrsv2_analysis	hipsparseZbsrsv2_analysis	
cusparseXbsrsv2_zeroPivot	hipsparseXbsrsv2_zeroPivot	
cusparseSbsrsv2_solve	hipsparseSbsrsv2_solve	
cusparseDbsrsv2_solve	hipsparseDbsrsv2_solve	
cusparseCbsrsv2_solve	hipsparseCbsrsv2_solve	
cusparseZbsrsv2_solve	hipsparseZbsrsv2_solve	
cusparseScsrsv_analysis		
cusparseDcsrsv_analysis		
cusparseCcsrsv_analysis		
cusparseZcsrsv_analysis		
cusparseCsrsv_analysisEx		8
cusparseScsrsv_solve		
cusparseDcsrsv_solve		
cusparseCcsrsv_solve		
cusparseZcsrsv_solve		

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cusparseCsrsv_solveEx		8
cusparseScsrsv2_bufferSize	hipsparseScsrsv2_bufferSize	
cusparseScsrsv2_bufferSizeExt	hipsparseScsrsv2_bufferSizeExt	
cusparseDcsrsv2_bufferSize	hipsparseDcsrsv2_bufferSize	
cusparseDcsrsv2_bufferSizeExt	hipsparseDcsrsv2_bufferSizeExt	
cusparseCcsrsv2_bufferSize	hipsparseCcsrsv2_bufferSize	
cusparseCcsrsv2_bufferSizeExt	hipsparseCcsrsv2_bufferSizeExt	
cusparseZcsrsv2_bufferSize	hipsparseZcsrsv2_bufferSize	
cusparseZcsrsv2_bufferSizeExt	hipsparseZcsrsv2_bufferSizeExt	
cusparseScsrsv2_analysis	hipsparseScsrsv2_analysis	
cusparseDcsrsv2_analysis	hipsparseDcsrsv2_analysis	
cusparseCcsrsv2_analysis	hipsparseCcsrsv2_analysis	
cusparseZcsrsv2_analysis	hipsparseZcsrsv2_analysis	
cusparseScsrsv2_solve	hipsparseScsrsv2_solve	
cusparseDcsrsv2_solve	hipsparseDcsrsv2_solve	
cusparseCcsrsv2_solve	hipsparseCcsrsv2_solve	
cusparseZcsrsv2_solve	hipsparseZcsrsv2_solve	
cusparseXcsrsv2_zeroPivot	hipsparseXcsrsv2_zeroPivot	
cusparseShybmv	hipsparseShybmv	
cusparseDhybmv	hipsparseDhybmv	
cusparseChybmv	hipsparseChybmv	
cusparseZhybmv	hipsparseZhybmv	
cusparseShybsv_analysis		
cusparseDhybsv_analysis		
cusparseChybsv_analysis		
cusparseZhybsv_analysis		
cusparseShybsv_solve		
cusparseDhybsv_solve		

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5.6.5 Level3 支持函数列表

CUDA	HIP	CUDA version*
cusparseDcsrmm	hipsparseDcsrmm	
cusparseCcsrmm	hipsparseCcsrmm	
cusparseZcsrmm	hipsparseZcsrmm	
cusparseScsrmm2	hipsparseScsrmm2	
cusparseDcsrmm2	hipsparseDcsrmm2	
cusparseCcsrmm2	hipsparseCcsrmm2	
cusparseZcsrmm2	hipsparseZcsrmm2	
cusparseScsrsm_analysis		
cusparseDcsrsm_analysis		
cusparseCcsrsm_analysis		
cusparseZcsrsm_analysis		
cusparseScsrsm_solve		
cusparseDcsrsm_solve		
cusparseCcsrsm_solve		
cusparseZcsrsm_solve		
cusparseScsrsm2_bufferSizeExt	hipsparseScsrsm2_bufferSizeExt	9.2
cusparseDcsrsm2_bufferSizeExt	hipsparseDcsrsm2_bufferSizeExt	9.2
cusparseCcsrsm2_bufferSizeExt	hipsparseCcsrsm2_bufferSizeExt	9.2
cusparseZcsrsm2_bufferSizeExt	hipsparseZcsrsm2_bufferSizeExt	9.2
cusparseScsrsm2_analysis	hipsparseScsrsm2_analysis	9.2
cusparseDcsrsm2_analysis	hipsparseDcsrsm2_analysis	9.2
cusparseCcsrsm2_analysis	hipsparseCcsrsm2_analysis	9.2
cusparseZcsrsm2_analysis	hipsparseZcsrsm2_analysis	9.2
cusparseScsrsm2_solve	hipsparseScsrsm2_solve	9.2
cusparseDcsrsm2_solve	hipsparseDcsrsm2_solve	9.2

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cusparseCcsrsm2_solve	hipsparseCcsrsm2_solve	9.2
cusparseZcsrsm2_solve	hipsparseZcsrsm2_solve	9.2
cusparseXcsrsm2_zeroPivot	hipsparseXcsrsm2_zeroPivot	9.2
cusparseSbsrmm	hipsparseSbsrmm	
cusparseDbsrmm	hipsparseDbsrmm	
cusparseCbsrmm	hipsparseCbsrmm	
cusparseZbsrmm	hipsparseZbsrmm	
cusparseSbsrsm2_bufferSize	hipsparseSbsrsm2_bufferSize	
cusparseSbsrsm2_bufferSizeExt		
cusparseDbsrsm2_bufferSize	hipsparseDbsrsm2_bufferSize	
cusparseDbsrsm2_bufferSizeExt		
cusparseCbsrsm2_bufferSize	hipsparseCbsrsm2_bufferSize	
cusparseCbsrsm2_bufferSizeExt		
cusparseZbsrsm2_bufferSize	hipsparseZbsrsm2_bufferSize	
cusparseZbsrsm2_bufferSizeExt		
cusparseSbsrsm2_analysis	hipsparseSbsrsm2_analysis	
cusparseDbsrsm2_analysis	hipsparseDbsrsm2_analysis	
cusparseCbsrsm2_analysis	hipsparseCbsrsm2_analysis	
cusparseZbsrsm2_analysis	hipsparseZbsrsm2_analysis	
cusparseSbsrsm2_solve	hipsparseSbsrsm2_solve	
cusparseDbsrsm2_solve	hipsparseDbsrsm2_solve	
cusparseCbsrsm2_solve	hipsparseCbsrsm2_solve	
cusparseZbsrsm2_solve	hipsparseZbsrsm2_solve	
cusparseXbsrsm2_zeroPivot	hipsparseXcsrsm2_zeroPivot	
cusparseSgemmi	hipsparseSgemmi	8
cusparseDgemmi	hipsparseDgemmi	8
cusparseCgemmi	hipsparseCgemmi	8
cusparseZgemmi	hipsparseZgemmi	8

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5.6.6 其他函数列表

CUDA	HIP	CUDA version *
cusparseXcsrgeamNnz	hipsparseXcsrgeamNnz	
cusparseScsrgeam	hipsparseScsrgeam	
cusparseDcsrgeam	hipsparseDcsrgeam	
cusparseCcsrgeam	hipsparseCcsrgeam	
cusparseZcsrgeam	hipsparseZcsrgeam	
cusparseXcsrgeam2Nnz	hipsparseXcsrgeam2Nnz	9.2
cusparseScsrgeam2	hipsparseScsrgeam2	9.2
cusparseDcsrgeam2	hipsparseDcsrgeam2	9.2
cusparseCcsrgeam2	hipsparseCcsrgeam2	9.2
cusparseZcsrgeam2	hipsparseZcsrgeam2	9.2
cusparseScsrgeam2_bufferSizeExt	hipsparseScsrgeam2_bufferSizeExt	9.2
cusparseDcsrgeam2_bufferSizeExt	hipsparseDcsrgeam2_bufferSizeExt	9.2
cusparseCcsrgeam2_bufferSizeExt	hipsparseCcsrgeam2_bufferSizeExt	9.2
cusparseZcsrgeam2_bufferSizeExt	hipsparseZcsrgeam2_bufferSizeExt	9.2
cusparseXcsrgemmNnz	hipsparseXcsrgemmNnz	
cusparseScsrgemm	hipsparseScsrgemm	
cusparseDcsrgemm	hipsparseDcsrgemm	
cusparseCcsrgemm	hipsparseCcsrgemm	
cusparseZcsrgemm	hipsparseZcsrgemm	
cusparseXcsrgemm2Nnz	hipsparseXcsrgemm2Nnz	
cusparseScsrgemm2	hipsparseScsrgemm2	
cusparseDcsrgemm2	hipsparseDcsrgemm2	
cusparseCcsrgemm2	hipsparseCcsrgemm2	
cusparseZcsrgemm2	hipsparseZcsrgemm2	

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cusparseScsrgemm2_bufferSizeExt	hipsparseScsrgemm2_bufferSizeExt	
cusparseDcsrgemm2_bufferSizeExt	hipsparseDcsrgemm2_bufferSizeExt	
cusparseCcsrgemm2_bufferSizeExt	hipsparseCcsrgemm2_bufferSizeExt	
cusparseZcsrgemm2_bufferSizeExt	hipsparseZcsrgemm2_bufferSizeExt	

5.6.7 预设条件函数接口

乔里斯基分解

CUDA	HIP	CUDA version*
cusparseScsric0		
cusparseDcsric0		
cusparseCcsric0		
cusparseZcsric0		
cusparseScsric02_bufferSize	hipsparseScsric02_bufferSize	
cusparseScsric02_bufferSizeExt	hipsparseScsric02_bufferSizeExt	
cusparseDcsric02_bufferSize	hipsparseDcsric02_bufferSize	
cusparseDcsric02_bufferSizeExt	hipsparseDcsric02_bufferSizeExt	
cusparseCcsric02_bufferSize	hipsparseCcsric02_bufferSize	
cusparseCcsric02_bufferSizeExt	hipsparseCcsric02_bufferSizeExt	
cusparseZcsric02_bufferSize	hipsparseZcsric02_bufferSize	
cusparseZcsric02_bufferSizeExt	hipsparseZcsric02_bufferSizeExt	
cusparseScsric02_analysis	hipsparseScsric02_analysis	
cusparseDcsric02_analysis	hipsparseDcsric02_analysis	
cusparseCcsric02_analysis	hipsparseCcsric02_analysis	
cusparseZcsric02_analysis	hipsparseZcsric02_analysis	

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cusparseScsric02	hipsparseScsric02	
cusparseDcsric02	hipsparseDcsric02	
cusparseCcsric02	hipsparseCcsric02	
cusparseZcsric02	hipsparseZcsric02	
cusparseXcsric02_zeroPivot	hipsparseXcsric02_zeroPivot	
cusparseSbsric02_bufferSize	hipsparseSbsric02_bufferSize	
cusparseSbsric02_bufferSizeExt		
cusparseDbsric02_bufferSize	hipsparseDbsric02_bufferSize	
cusparseDbsric02_bufferSizeExt		
cusparseCbsric02_bufferSize	hipsparseCbsric02_bufferSize	
cusparseCbsric02_bufferSizeExt		
cusparseZbsric02_bufferSize	hipsparseZbsric02_bufferSize	
cusparseZbsric02_bufferSizeExt		
cusparseSbsric02_analysis	hipsparseSbsric02_analysis	
cusparseDbsric02_analysis	hipsparseDbsric02_analysis	
cusparseCbsric02_analysis	hipsparseCbsric02_analysis	
cusparseZbsric02_analysis	hipsparseZbsric02_analysis	
cusparseSbsric02	hipsparseSbsric02	
cusparseDbsric02	hipsparseDbsric02	
cusparseCbsric02	hipsparseCbsric02	
cusparseZbsric02	hipsparseZbsric02	
cusparseXbsric02_zeroPivot	hipsparseXbsric02_zeroPivot	

LU 分解

CUDA	HIP	CUDA version*
cusparseScsrilu0		
cusparseDcsrilu0		
cusparseCcsrilu0		

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cusparseZcsrilu0		
cusparseCsrilu0Ex		8
cusparseScsrilu02_numericBoost	hipsparseScsrilu02_numericBoost	
cusparseDcsrilu02_numericBoost	hipsparseDcsrilu02_numericBoost	
cusparseCcsrilu02_numericBoost	hipsparseCcsrilu02_numericBoost	
cusparseZcsrilu02_numericBoost	hipsparseZcsrilu02_numericBoost	
cusparseXcsrilu02_zeroPivot	hipsparseXcsrilu02_zeroPivot	
cusparseScsrilu02_bufferSize	hipsparseScsrilu02_bufferSize	
cusparseScsrilu02_bufferSizeExt	hipsparseScsrilu02_bufferSizeExt	
cusparseDcsrilu02_bufferSize	hipsparseDcsrilu02_bufferSize	
cusparseDcsrilu02_bufferSizeExt	hipsparseDcsrilu02_bufferSizeExt	
cusparseCcsrilu02_bufferSize	hipsparseCcsrilu02_bufferSize	
cusparseCcsrilu02_bufferSizeExt	hipsparseCcsrilu02_bufferSizeExt	
cusparseZcsrilu02_bufferSize	hipsparseZcsrilu02_bufferSize	
cusparseZcsrilu02_bufferSizeExt	hipsparseZcsrilu02_bufferSizeExt	
cusparseScsrilu02_analysis	hipsparseScsrilu02_analysis	
cusparseDcsrilu02_analysis	hipsparseDcsrilu02_analysis	
cusparseCcsrilu02_analysis	hipsparseCcsrilu02_analysis	
cusparseZcsrilu02_analysis	hipsparseZcsrilu02_analysis	
cusparseScsrilu02	hipsparseScsrilu02	
cusparseDcsrilu02	hipsparseDcsrilu02	
cusparseCcsrilu02	hipsparseCcsrilu02	
cusparseZcsrilu02	hipsparseZcsrilu02	
cusparseXbsric02_zeroPivot	hipsparseXcsrilu02_zeroPivot	
cusparseSbsrilu02_numericBoost	hipsparseSbsrilu02_numericBoost	
cusparseDbsrilu02_numericBoost	hipsparseDbsrilu02_numericBoost	
cusparseCbsrilu02_numericBoost	hipsparseCbsrilu02_numericBoost	
cusparseZbsrilu02_numericBoost	hipsparseZbsrilu02_numericBoost	

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cusparseSbsrilu02_bufferSize	hipsparseSbsrilu02_bufferSize	
cusparseSbsrilu02_bufferSizeExt		
cusparseDbsrilu02_bufferSize	hipsparseDbsrilu02_bufferSize	
cusparseDbsrilu02_bufferSizeExt		
cusparseCbsrilu02_bufferSize	hipsparseCbsrilu02_bufferSize	
cusparseCbsrilu02_bufferSizeExt		
cusparseZbsrilu02_bufferSize	hipsparseZbsrilu02_bufferSize	
cusparseZbsrilu02_bufferSizeExt		
cusparseSbsrilu02_analysis	hipsparseSbsrilu02_analysis	
cusparseDbsrilu02_analysis	hipsparseDbsrilu02_analysis	
cusparseCbsrilu02_analysis	hipsparseCbsrilu02_analysis	
cusparseZbsrilu02_analysis	hipsparseZbsrilu02_analysis	
cusparseSbsrilu02	hipsparseSbsrilu02	
cusparseDbsrilu02	hipsparseDbsrilu02	
cusparseCbsrilu02	hipsparseCbsrilu02	
cusparseZbsrilu02	hipsparseZbsrilu02	
cusparseXbsrilu02_zeroPivot	hipsparseXbsrilu02_zeroPivot	

三对角矩阵求解

CUDA	HIP	CUDA version *
cusparseSgtsv		
cusparseDgtsv		
cusparseCgtsv		
cusparseZgtsv		
cusparseSgtsv_nopivot		
cusparseDgtsv_nopivot		
cusparseCgtsv_nopivot		

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cusparseZgtsv_nopivot		
cusparseSgtsv2_bufferSizeExt	hipsparseSgtsv2_bufferSizeExt	9
cusparseDgtsv2_bufferSizeExt	hipsparseDgtsv2_bufferSizeExt	9
cusparseCgtsv2_bufferSizeExt	hipsparseCgtsv2_bufferSizeExt	9
cusparseZgtsv2_bufferSizeExt	hipsparseZgtsv2_bufferSizeExt	9
cusparseSgtsv2	hipsparseSgtsv2	9
cusparseDgtsv2	hipsparseDgtsv2	9
cusparseCgtsv2	hipsparseCgtsv2	9
cusparseZgtsv2	hipsparseZgtsv2	9
cusparseSgtsv2_nopivot_bufferSizeExt	hipsparseSgtsv2_nopivot_bufferSizeExt	9
cusparseDgtsv2_nopivot_bufferSizeExt	hipsparseDgtsv2_nopivot_bufferSizeExt	9
cusparseCgtsv2_nopivot_bufferSizeExt	hipsparseCgtsv2_nopivot_bufferSizeExt	9
cusparseZgtsv2_nopivot_bufferSizeExt	hipsparseZgtsv2_nopivot_bufferSizeExt	9
cusparseSgtsv2_nopivot	hipsparseSgtsv2_nopivot	9
cusparseDgtsv2_nopivot	hipsparseDgtsv2_nopivot	9
cusparseCgtsv2_nopivot	hipsparseCgtsv2_nopivot	9
cusparseZgtsv2_nopivot	hipsparseZgtsv2_nopivot	9

CUDA	HIP	CUDA version*
cusparseSgtsvStridedBatch		
cusparseDgtsvStridedBatch		

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cusparseCgtsvStridedBatch		
cusparseZgtsvStridedBatch		
cusparseSgtsv2StridedBatch_bufferSizeExt	hipsparseSgtsv2StridedBatch_bufferSizeExt	9
cusparseDgtsv2StridedBatch_bufferSizeExt	hipsparseDgtsv2StridedBatch_bufferSizeExt	9
cusparseCgtsv2StridedBatch_bufferSizeExt	hipsparseCgtsv2StridedBatch_bufferSizeExt	9
cusparseZgtsv2StridedBatch_bufferSizeExt	hipsparseZgtsv2StridedBatch_bufferSizeExt	9
cusparseSgtsv2StridedBatch	hipsparseSgtsv2StridedBatch	9
cusparseDgtsv2StridedBatch	hipsparseDgtsv2StridedBatch	9
cusparseCgtsv2StridedBatch	hipsparseCgtsv2StridedBatch	9
cusparseZgtsv2StridedBatch	hipsparseZgtsv2StridedBatch	9
cusparseSgtsvInterleavedBatch_bufferSizeExt	hipsparseSgtsvInterleavedBatch_bufferSizeExt	9.2
cusparseDgtsvInterleavedBatch_bufferSizeExt	hipsparseDgtsvInterleavedBatch_bufferSizeExt	9.2
cusparseCgtsvInterleavedBatch_bufferSizeExt	hipsparseCgtsvInterleavedBatch_bufferSizeExt	9.2
cusparseZgtsvInterleavedBatch_bufferSizeExt	hipsparseZgtsvInterleavedBatch_bufferSizeExt	9.2
cusparseSgtsvInterleavedBatch	hipsparseSgtsvInterleavedBatch	9.2
cusparseDgtsvInterleavedBatch	hipsparseDgtsvInterleavedBatch	9.2
cusparseCgtsvInterleavedBatch	hipsparseCgtsvInterleavedBatch	9.2
cusparseZgtsvInterleavedBatch	hipsparseZgtsvInterleavedBatch	9.2

PentadiagonalSolve

CUDA	HIP	CUDA version*

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cusparseSgpsvInterleavedBatch_bufferSizeExt	hipsparseSgpsvInterleavedBatch_bufferSizeExt	9.2
cusparseDgpsvInterleavedBatch_bufferSizeExt	hipsparseDgpsvInterleavedBatch_bufferSizeExt	9.2
cusparseCgpsvInterleavedBatch_bufferSizeExt	hipsparseCgpsvInterleavedBatch_bufferSizeExt	9.2
cusparseZgpsvInterleavedBatch_bufferSizeExt	hipsparseZgpsvInterleavedBatch_bufferSizeExt	9.2
cusparseSgpsvInterleavedBatch	hipsparseSgpsvInterleavedBatch	9.2
cusparseDgpsvInterleavedBatch	hipsparseDgpsvInterleavedBatch	9.2
cusparseCgpsvInterleavedBatch	hipsparseCgpsvInterleavedBatch	9.2
cusparseZgpsvInterleavedBatch	hipsparseZgpsvInterleavedBatch	9.2

矩阵重排

CUDA	HIP	CUDA version*
cusparseScsrcolor	hipsparseScsrcolor	9.2
cusparseDcsrcolor	hipsparseDcsrcolor	9.2
cusparseCcsrcolor	hipsparseCcsrcolor	9.2
cusparseZcsrcolor	hipsparseZcsrcolor	

格式转换

CUDA	HIP	CUDA version*
cusparseSbsr2csr	hipsparseSbsr2csr	
cusparseDbsr2csr	hipsparseDbsr2csr	
cusparseCbsr2csr	hipsparseCbsr2csr	

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cusparseZbsr2csr	hipsparseZbsr2csr	
cusparseSgebsr2gebsc_bufferSize	hipsparseSgebsr2gebsc_bufferSize	
cusparseSgebsr2gebsc_bufferSizeExt		
cusparseDgebsr2gebsc_bufferSize	hipsparseDgebsr2gebsc_bufferSize	
cusparseDgebsr2gebsc_bufferSizeExt		
cusparseCgebsr2gebsc_bufferSize	hipsparseCgebsr2gebsc_bufferSize	
cusparseCgebsr2gebsc_bufferSizeExt		
cusparseZgebsr2gebsc_bufferSize	hipsparseZgebsr2gebsc_bufferSize	
cusparseZgebsr2gebsc_bufferSizeExt		
cusparseSgebsr2gebsc	hipsparseSgebsr2gebsc	
cusparseDgebsr2gebsc	hipsparseDgebsr2gebsc	
cusparseCgebsr2gebsc	hipsparseCgebsr2gebsc	
cusparseZgebsr2gebsc	hipsparseZgebsr2gebsc	
cusparseSgebsr2gebsr_bufferSize	hipsparseSgebsr2gebsr_bufferSize	
cusparseSgebsr2gebsr_bufferSizeExt		
cusparseDgebsr2gebsr_bufferSize	hipsparseDgebsr2gebsr_bufferSize	
cusparseDgebsr2gebsr_bufferSizeExt		
cusparseCgebsr2gebsr_bufferSize	hipsparseCgebsr2gebs	

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	r_bufferSize	
cusparseCgebsr2gebsr_bufferSizeExt		
cusparseZgebsr2gebsr_bufferSize	hipsparseZgebsr2gebsr_bufferSize	
cusparseZgebsr2gebsr_bufferSizeExt		
cusparseXgebsr2gebsrNnz	hipsparseXgebsr2gebsrNnz	
cusparseSgebsr2gebsr	hipsparseSgebsr2gebsr	
cusparseDgebsr2gebsr	hipsparseDgebsr2gebsr	
cusparseCgebsr2gebsr	hipsparseCgebsr2gebsr	
cusparseZgebsr2gebsr	hipsparseZgebsr2gebsr	
cusparseXgebsr2csr	hipsparseXgebsr2csr	
cusparseSgebsr2csr	hipsparseSgebsr2csr	
cusparseDgebsr2csr	hipsparseDgebsr2csr	
cusparseCgebsr2csr	hipsparseCgebsr2csr	
cusparseZgebsr2csr	hipsparseZgebsr2csr	
cusparseScsr2gebsr_bufferSize	hipsparseScsr2gebsr_bufferSize	
cusparseScsr2gebsr_bufferSizeExt		
cusparseDcsr2gebsr_bufferSize	hipsparseDcsr2gebsr_bufferSize	
cusparseDcsr2gebsr_bufferSizeExt		
cusparseCcsr2gebsr_bufferSize	hipsparseCcsr2gebsr_bufferSize	

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cusparseCcsr2gebsr_bufferSizeExt		
cusparseZcsr2gebsr_bufferSize	hipsparseZcsr2gebsr_bufferSize	
cusparseZcsr2gebsr_bufferSizeExt		
cusparseXcsr2gebsrNnz	hipsparseXcsr2gebsrNnz	
cusparseScsr2gebsr	hipsparseScsr2gebsr	
cusparseDcsr2gebsr	hipsparseDcsr2gebsr	
cusparseCcsr2gebsr	hipsparseCcsr2gebsr	
cusparseZcsr2gebsr	hipsparseZcsr2gebsr	
cusparseXcoo2csr	hipsparseXcoo2csr	
cusparseScsc2dense	hipsparseScsc2dense	
cusparseDcsc2dense	hipsparseDcsc2dense	
cusparseCcsc2dense	hipsparseCcsc2dense	
cusparseZcsc2dense	hipsparseZcsc2dense	
cusparseScsc2hyb		
cusparseDcsc2hyb		
cusparseCcsc2hyb		
cusparseZcsc2hyb		
cusparseXcsr2bsrNnz	hipsparseXcsr2bsrNnz	
cusparseScsr2bsr	hipsparseScsr2bsr	
cusparseDcsr2bsr	hipsparseDcsr2bsr	
cusparseCcsr2bsr	hipsparseCcsr2bsr	
cusparseZcsr2bsr	hipsparseZcsr2bsr	
cusparseXcsr2coo	hipsparseXcsr2coo	
cusparseScsr2csc	hipsparseScsr2csc	
cusparseDcsr2csc	hipsparseDcsr2csc	
cusparseCcsr2csc	hipsparseCcsr2csc	

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cusparseZcsr2csc	hipsparseZcsr2csc	
cusparseCsr2cscEx		8
cusparseCsr2cscEx2		10.1
cusparseCsr2cscEx2_bufferSize		10.1
cusparseScsr2dense	hipsparseScsr2dense	
cusparseDcsr2dense	hipsparseScsr2dense	
cusparseCcsr2dense	hipsparseCcsr2dense	
cusparseZcsr2dense	hipsparseZcsr2dense	
cusparseScsr2csr_compress	hipsparseScsr2csr_compress	8
cusparseDcsr2csr_compress	hipsparseDcsr2csr_compress	8
cusparseCcsr2csr_compress	hipsparseCcsr2csr_compress	8
cusparseZcsr2csr_compress	hipsparseZcsr2csr_compress	8
cusparseScsr2hyb	hipsparseScsr2hyb	
cusparseDcsr2hyb	hipsparseDcsr2hyb	
cusparseCcsr2hyb	hipsparseCcsr2hyb	
cusparseZcsr2hyb	hipsparseZcsr2hyb	
cusparseSdense2csc	hipsparseSdense2csc	
cusparseDdense2csc	hipsparseDdense2csc	
cusparseCdense2csc	hipsparseCdense2csc	
cusparseZdense2csc	hipsparseZdense2csc	
cusparseSdense2csr	hipsparseSdense2csr	
cusparseDdense2csr	hipsparseDdense2csr	
cusparseCdense2csr	hipsparseCdense2csr	
cusparseZdense2csr	hipsparseZdense2csr	

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cusparseSdense2hyb		
cusparseDdense2hyb		
cusparseCdense2hyb		
cusparseZdense2hyb		
cusparseShyb2csc		
cusparseDhyb2csc		
cusparseChyb2csc		
cusparseZhyb2csc		
cusparseShyb2csr	hipsparseShyb2csr	
cusparseDhyb2csr	hipsparseDhyb2csr	
cusparseChyb2csr	hipsparseChyb2csr	
cusparseZhyb2csr	hipsparseZhyb2csr	
cusparseShyb2dense		
cusparseDhyb2dense		
cusparseChyb2dense		
cusparseZhyb2dense		
cusparseSnnz	cusparseSnnz	
cusparseDnnz	cusparseDnnz	
cusparseCnnz	cusparseCnnz	
cusparseZnnz	cusparseZnnz	
cusparseCreateIdentityPermutation	hipsparseCreateIdentityPermutation	
cusparseXcoosort_bufferSizeExt	hipsparseXcoosort_bufferSizeExt	
cusparseXcoosortByRow	hipsparseXcoosortByRow	
cusparseXcoosortByColumn	hipsparseXcoosortByColumn	

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cusparseXcsrsort_bufferSizeExt	hipsparseXcsrsort_bufferSizeExt	
cusparseXcsrsort	hipsparseXcsrsort	
cusparseXcscsort_bufferSizeExt	hipsparseXcscsort_bufferSizeExt	
cusparseXcscsort	hipsparseXcscsort	
cusparseCreateCsru2csrInfo	hipsparseCreateCsru2csrInfo	
cusparseDestroyCsru2csrInfo	hipsparseDestroyCsru2csrInfo	
cusparseScsru2csr_bufferSizeExt	hipsparseScsru2csr_bufferSizeExt	
cusparseDcsru2csr_bufferSizeExt	hipsparseDcsru2csr_bufferSizeExt	
cusparseCcsru2csr_bufferSizeExt	hipsparseCcsru2csr_bufferSizeExt	
cusparseZcsru2csr_bufferSizeExt	hipsparseZcsru2csr_bufferSizeExt	
cusparseScsru2csr	hipsparseScsru2csr	
cusparseDcsru2csr	hipsparseDcsru2csr	
cusparseCcsru2csr	hipsparseCcsru2csr	
cusparseZcsru2csr	hipsparseZcsru2csr	
cusparseScsr2csru	hipsparseScsr2csru	
cusparseDcsr2csru	hipsparseDcsr2csru	
cusparseCcsr2csru	hipsparseCcsr2csru	
cusparseZcsr2csru	hipsparseZcsr2csru	
cusparseHpruneDense2csr		9
cusparseSpruneDense2csr	hipsparseSpruneDense	9

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	2csr	
cusparseDpruneDense2csr	hipsparseDpruneDense2csr	9
cusparseHpruneDense2csr_bufferSizeExt	hipsparseSpruneDense2csr_bufferSizeExt	9
cusparseSpruneDense2csr_bufferSizeExt	hipsparseSpruneDense2csr_bufferSizeExt	9
cusparseDpruneDense2csr_bufferSizeExt	hipsparseDpruneDense2csr_bufferSizeExt	9
cusparseHpruneDense2csrNnz		9
cusparseSpruneDense2csrNnz	hipsparseSpruneDense2csrNnz	9
cusparseDpruneDense2csrNnz	hipsparseDpruneDense2csrNnz	9
cusparseHpruneCsr2csr		9
cusparseSpruneCsr2csr	hipsparseSpruneCsr2csr	9
cusparseDpruneCsr2csr	hipsparseDpruneCsr2csr	9
cusparseHpruneCsr2csr_bufferSizeExt		9
cusparseSpruneCsr2csr_bufferSizeExt	hipsparseSpruneCsr2csr_bufferSizeExt	9
cusparseDpruneCsr2csr_bufferSizeExt	hipsparseDpruneCsr2csr_bufferSizeExt	9
cusparseHpruneCsr2csrNnz		9
cusparseSpruneCsr2csrNnz	hipsparseSpruneCsr2csrNnz	9
cusparseDpruneCsr2csrNnz	hipsparseDpruneCsr2c	9

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	srNnz	
cusparseHpruneDense2csrByPercentage		9
cusparseSpruneDense2csrByPercentage	hipsparseSpruneDense2csrByPercentage	9
cusparseDpruneDense2csrByPercentage	hipsparseDpruneDense2csrByPercentage	9
cusparseHpruneDense2csrByPercentage_bufferSizeExt		9
cusparseSpruneDense2csrByPercentage_bufferSizeExt	hipsparseSpruneDense2csrByPercentage_bufferSizeExt	9
cusparseDpruneDense2csrByPercentage_bufferSizeExt	hipsparseDpruneDense2csrByPercentage_bufferSizeExt	9
cusparseHpruneDense2csrNnzByPercentage		9
cusparseSpruneDense2csrNnzByPercentage	hipsparseSpruneDense2csrNnzByPercentage	9
cusparseDpruneDense2csrNnzByPercentage	hipsparseDpruneDense2csrNnzByPercentage	9
cusparseHpruneCsr2csrByPercentage		9
cusparseSpruneCsr2csrByPercentage	hipsparseSpruneCsr2csrByPercentage	9
cusparseDpruneCsr2csrByPercentage	hipsparseDpruneCsr2csrByPercentage	9
cusparseHpruneCsr2csrByPercentage_bufferSizeExt		9
cusparseSpruneCsr2csrByPercentage_buff	hipsparseSpruneCsr2c	9

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erSizeExt	srByPercentage_buffe rSizeExt	
cusparseDpruneCsr2csrByPercentage_buff erSizeExt	hipsparseDpruneCsr2c srByPercentage_buffe rSizeExt	9
cusparseHpruneCsr2csrNnzByPercentage		9
cusparseSpruneCsr2csrNnzByPercentage	hipsparseSpruneCsr2c srNnzByPercentage	9
cusparseDpruneCsr2csrNnzByPercentage	hipsparseDpruneCsr2c srNnzByPercentage	9
cusparseZnnz	hipsparseZnnz	
cusparseCnnz	hipsparseCnnz	
cusparseDnnz	hipsparseDnnz	
cusparseSnnz	hipsparseSnnz	
cusparseSnnz_compress	hipsparseSnnz_compr ess	8
cusparseDnnz_compress	hipsparseDnnz_compr ess	8
cusparseCnnz_compress	hipsparseCnnz_compr ess	8
cusparseZnnz_compress	hipsparseZnnz_compr ess	8

矩阵生成函数

CUDA	HIP	CUDA version*
cusparseCreateCoo	hipsparseCreateCoo	10.1
cusparseCreateCooAoS	hipsparseCreateCooAoS	10.1
cusparseCreateCsr	hipsparseCreateCsr	10.1

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cusparseDestroySpMat	hipsparseDestroySpMat	10.1
cusparseCooGet	hipsparseCooGet	10.1
cusparseCooAoSGet	hipsparseCooAoSGet	10.1
cusparseCsrGet	hipsparseCsrGet	10.1
cusparseSpMatGetFormat	hipsparseSpMatGetFormat	10.1
cusparseSpMatGetIndexBase	hipsparseSpMatGetIndexBase	10.1
cusparseSpMatGetValues	hipsparseSpMatGetValues	10.1
cusparseSpMatSetValues	hipsparseSpMatSetValues	10.1
cusparseSpMatGetStridedBatch	hipsparseSpMatGetStridedBatch	10.1
cusparseSpMatSetStridedBatch	hipsparseSpMatSetStridedBatch	10.1
cusparseSpMatGetNumBatches		10.1
cusparseSpMatSetNumBatches		10.1
cusparseCreateSpVec	hipsparseCreateSpVec	10.1
cusparseDestroySpVec	hipsparseDestroySpVec	10.1
cusparseSpVecGet	hipsparseSpVecGet	10.1
cusparseSpVecGetIndexBase	hipsparseSpVecGetIndexBase	10.1
cusparseSpVecGetValues	hipsparseSpVecGetValues	10.1
cusparseSpVecSetValues	hipsparseSpVecSetValues	10.1
cusparseCreateCoo	hipsparseCreateCoo	10.1
cusparseCreateCooAoS	hipsparseCreateCooAoS	10.1
cusparseCreateCsr	hipsparseCreateCsr	10.1

稠密矩阵和向量生成函数

CUDA	HIP	CUDA version*
cusparseCreateDnMat	hipsparseCreateDnMat	10.1
cusparseDestroyDnMat	hipsparseDestroyDnMat	10.1
cusparseDnMatGet	hipsparseDnMatGet	10.1
cusparseDnMatGetValues	hipsparseDnMatGetValues	10.1

cusparseDnMatSetValues	hipsparseDnMatSetValues	10.1
cusparseDnMatSetStridedBatch	hipsparseDnMatSetStridedBatch	10.1
cusparseDnMatGetStridedBatch	hipsparseDnMatGetStridedBatch	10.1
cusparseCreateDnVec	hipsparseCreateDnVec	10.1
cusparseDestroyDnVec	hipsparseDestroyDnVec	10.1
cusparseDnVecGet	hipsparseDnVecGet	10.1
cusparseDnVecGetValues	hipsparseDnVecGetValues	10.1
cusparseDnVecSetValues	hipsparseDnVecSetValues	10.1

SpMM 创建函数

CUDA	HIP	CUDA version*
cusparseSpMM	hipsparseSpMM	10.1
cusparseSpMM_bufferSize	hipsparseSpMM_bufferSize	

SpMV 创建函数

CUDA	HIP	CUDA version*
cusparseSpMV	hipsparseSpMV	10.1
cusparseSpMV_bufferSize	hipsparseSpMV_bufferSize	10.1

SpVV 创建函数

CUDA	HIP	CUDA version*
cusparseSpVV	hipsparseSpVV	
cusparseSpVV_bufferSize	hipsparseSpVV_bufferSize	

5.7 RAND

5.7.1 Data Types

type	CUDA	HIP	CUDA version *
enum	curandStatus_t	hiprandStatus_t	10.2
0	CURAND_STATUS_SUCC	HIPRAND_STATUS_SUCC	

	ESS	ESS	
1	CURAND_STATUS_VERSION_MISMATCH	HIPRAND_STATUS_VERSION_MISMATCH	
2	CURAND_STATUS_NOT_INITIALIZED	HIPRAND_STATUS_NOT_INITIALIZED	
3	CURAND_STATUS_ALLOCATION_FAILED	HIPRAND_STATUS_ALLOCATION_FAILED	
4	CURAND_STATUS_TYPE_ERROR	HIPRAND_STATUS_TYPE_ERROR	
5	CURAND_STATUS_OUT_OF_RANGE	HIPRAND_STATUS_OUT_OF_RANGE	
6	CURAND_STATUS_LENGTH_NOT_MULTIPLE	HIPRAND_STATUS_LENGTH_NOT_MULTIPLE	
7	CURAND_STATUS_DOUBLE_PRECISION_REQUIRED	HIPRAND_STATUS_DOUBLE_PRECISION_REQUIRED	
8	CURAND_STATUS_LAUNCH_FAILURE	HIPRAND_STATUS_LAUNCH_FAILURE	
9	CURAND_STATUS_PREEXISTING_FAILURE	HIPRAND_STATUS_PREEXISTING_FAILURE	
10	CURAND_STATUS_INITIALIZATION_FAILED	HIPRAND_STATUS_INITIALIZATION_FAILED	
11	CURAND_STATUS_ARCH_MISMATCH	HIPRAND_STATUS_ARCH_MISMATCH	
12	CURAND_STATUS_INTERNAL_ERROR	HIPRAND_STATUS_INTERNAL_ERROR	
enum	curandRngType_t	hiprandRngType_t	
0	CURAND_RNG_TEST	HIPRAND_RNG_TEST	
1	CURAND_RNG_PSEUDO	HIPRAND_RNG_PSEUDO_	

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	_DEFAULT	DEFAULT	
2	CURAND_RNG_PSEUDO_XORWOW	HIPRAND_RNG_PSEUDO_XORWOW	
3	CURAND_RNG_PSEUDO_MRG32K3A	HIPRAND_RNG_PSEUDO_MRG32K3A	
4	CURAND_RNG_PSEUDO_MTGP32	HIPRAND_RNG_PSEUDO_MTGP32	
5	CURAND_RNG_PSEUDO_MT19937	HIPRAND_RNG_PSEUDO_MT19937	
6	CURAND_RNG_PSEUDO_PHILOX4_32_10	HIPRAND_RNG_PSEUDO_PHILOX4_32_10	
7	CURAND_RNG_QUASI_DEFAULT	HIPRAND_RNG_QUASI_DEFAULT	
8	CURAND_RNG_QUASI_SOBOL32	HIPRAND_RNG_QUASI_SOBOL32	
9	CURAND_RNG_QUASI_SCRAMBLED_SOBOL32	HIPRAND_RNG_QUASI_SCRAMBLED_SOBOL32	
10	CURAND_RNG_QUASI_SOBOL64	HIPRAND_RNG_QUASI_SOBOL64	
11	CURAND_RNG_QUASI_SCRAMBLED_SOBOL64	HIPRAND_RNG_QUASI_SCRAMBLED_SOBOL64	
enum	curandOrdering_t		
0	CURAND_ORDERING_PSEUDO_BEST		
1	CURAND_ORDERING_PSEUDO_DEFAULT		
2	CURAND_ORDERING_PSEUDO_SEEDED		
3	CURAND_ORDERING_Q		

	UASI_DEFAULT		
enum	curandDirectionVectorSet_t		
0	CURAND_DIRECTION_VECTORS_32_JOEKUO6		
1	CURAND_SCRAMBLED_DIRECTION_VECTORS_32_JOEKUO6		
2	CURAND_DIRECTION_VECTORS_64_JOEKUO6		
3	CURAND_SCRAMBLED_DIRECTION_VECTORS_64_JOEKUO6		
enum	curandMethod_t		
0	CURAND_CHOOSE_BEST		
1	CURAND_ITR		
2	CURAND_KNUTH		
3	CURAND_HITR		
4	CURAND_M1		
5	CURAND_M2		
6	CURAND_BINARY_SEARCH		
7	CURAND_DISCRETE_GAUSS		
8	CURAND_REJECTION		
9	CURAND_DEVICE_API		
10	CURAND_FAST_REJECTION		

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11	CURAND_3RD		
12	CURAND_DEFINITION		
13	CURAND_POISSON		
struct	curandGenerator_t	hiprandGenerator_t	
struct	curandDistribution_t		
struct	curandDistributionShift_t		
struct	curandDistributionM2Shift_t		
struct	curandHistogramM2_t		
struct	curandHistogramM2K_t		
struct	curandHistogramM2V_t		
struct	curandDiscreteDistribution_t	hiprandDiscreteDistribution_t	
	curandDirectionVectors32_t	hiprandDirectionVectors32_t	
struct	curandState_t	hiprandState	
struct	curandStateTest_t		
struct	curandStateXORWOW_t	hiprandStateXORWOW	
struct	curandStateMRG32k3a_t	hiprandStateMRG32k3a	
struct	curandStateSobol32_t	hiprandStateSobol32	
struct	curandStateScrambledSobol32_t	hiprandStateScrambledSobol32	
struct	curandStateSobol64_t	hiprandStateSobol64	
struct	curandStateScrambledSobol64_t	hiprandStateScrambledSobol64	
struct	curandStateMtgp32_t	hiprandStateMtgp32	
		hiprandStatePhilox4_32_10	
struct	normal_args_t		
struct	normal_args_double_t		

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struct	mtgp32_params_fast_t	mtgp32_fast_param_t	
struct	mtgp32_kernel_params_t	mtgp32_kernel_params_t	

5.7.2 API Functions

CUDA	HIP	CUDA version *
curandCreateGenerator	hiprandCreateGenerator	10.2
curandCreateGeneratorHost	hiprandCreateGeneratorHost	
curandDestroyGenerator	hiprandDestroyGenerator	
curandGetVersion	hiprandGetVersion	
curandGetProperty		
curandSetStream	hiprandSetStream	
curandSetPseudoRandomGeneratorSeed	hiprandSetPseudoRandomGeneratorSeed	
curandSetGeneratorOffset	hiprandSetGeneratorOffset	
curandSetGeneratorOrdering		
curandSetQuasiRandomGeneratorDimensions	hiprandSetQuasiRandomGeneratorDimensions	
curandGenerate	hiprandGenerate	
	hiprandGenerateChar	
	hiprandGenerateShort	
curandGenerateLongLong		
curandGenerateUniform	hiprandGenerateUniform	
curandGenerateUniformDouble	hiprandGenerateUniformDouble	
	hiprandGenerateUniformHalf	
curandGenerateNormal	hiprandGenerateNormal	
curandGenerateNormalDouble	hiprandGenerateNormalDouble	
	hiprandGenerateNormalHalf	

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curandGenerateLogNormal	hiprandGenerateLogNormal	
curandGenerateLogNormalDouble	hiprandGenerateLogNormalDouble	
	hiprandGenerateLogNormalHalf	
curandCreatePoissonDistribution	hiprandCreatePoissonDistribution	
curandDestroyDistribution	hiprandDestroyDistribution	
curandGeneratePoisson	hiprandGeneratePoisson	
curandGeneratePoissonMethod		
curandGenerateBinomial		
curandGenerateBinomialMethod		
curandGenerateSeeds	hiprandGenerateSeeds	
curandGetDirectionVectors32		
curandGetScrambleConstants32		
curandGetDirectionVectors64		
curandGetScrambleConstants64		
curand_init	hiprand_init	
curand	hiprand	
curand4	hiprand4	
curand_MRG32k3a		
curand_discrete	hiprand_discrete	
curand_discrete4	hiprand_discrete4	
curand_log_normal	hiprand_log_normal	
curand_log_normal2	hiprand_log_normal2	
curand_log_normal4	hiprand_log_normal4	
curand_log_normal_double	hiprand_log_normal_double	
curand_log_normal2_double	hiprand_log_normal2_double	
curand_log_normal4_double	hiprand_log_normal4_double	
curandMakeMTGP32Constants	hiprandMakeMTGP32Constants	

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curandMakeMTGP32KernelState	hiprandMakeMTGP32KernelState	
curand_mtgp32_specific		
curand_mtgp32_single		
curand_mtgp32_single_specific		
curand_normal	hiprand_normal	
curand_normal2	hiprand_normal2	
curand_normal4	hiprand_normal4	
curand_normal_double	hiprand_normal_double	
curand_normal2_double	hiprand_normal2_double	
curand_normal4_double	hiprand_normal4_double	
curand_uniform	hiprand_uniform	
curand_uniform_double	hiprand_uniform_double	
curand_uniform4	hiprand_uniform4	
	hiprand_uniform2_double	
curand_uniform4_double	hiprand_uniform4_double	
skipahead	skipahead	
skipaheadsequence	skipaheadsequence	
skipahead_subsequence	skipahead_subsequence	
curand_poisson	hiprand_poisson	
curand_poisson4	hiprand_poisson4	
curand_poisson_knuth		
curand_poisson_knuth4		
curand_poisson_gammainc		
curand_poisson_gammainc4		
curand_poisson_from_normal		
curand_M2_double		
curand_M2_double4		
curand_binary_search_double		

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5.8 CUB

hipCUB 在 rocPRIM 的基础上实现的，主要功能接口与 CUB 一致。hipCUB 与 CUB 的后端实现不一样，CUB 内部的 Agent、block/specialization 及 device/dispatch_*相关内容 hipCUB 均不支持。

5.8.1 Warp-wide "collective" primitives

CUDA	HIP	CUDA version*
cub::WarpExchange	hipcub::WarpExchange	
cub::WarpExchange::BlockedToStriped	hipcub::WarpExchange::BlockedToStriped	
cub::WarpExchange::StripedToBlocked	hipcub::WarpExchange::StripedToBlocked	
cub::WarpExchange::ScatterToStriped	hipcub::WarpExchange::ScatterToStriped	
cub::WarpLoad	hipcub::WarpLoad	
cub::WarpLoad::TempStorage	hipcub::WarpLoad::TempStorage	
cub::WarpLoad::Load	hipcub::WarpLoad::Load	
cub::WarpMergeSort	hipcub::WarpMergeSort	
cub::WarpMergeSort::get_member_mask	hipcub::WarpMergeSort::get_member_mask	
cub::WarpReduce	hipcub::WarpReduce	
cub::WarpReduce::TempStorage	hipcub::WarpReduce::TempStorage	
cub::WarpReduce::Sum	hipcub::WarpReduce::Sum	
cub::WarpReduce::HeadSegmentedSum	hipcub::WarpReduce::HeadSegmentedSum	
cub::WarpReduce::TailSegmentedSum	hipcub::WarpReduce::TailSegmentedSum	

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cub::WarpReduce::Reduce	hipcub::WarpReduce::Reduce	
cub::WarpReduce::HeadSegmentedReduce	hipcub::WarpReduce::HeadSegmentedReduce	
cub::WarpReduce::TailSegmentedReduce	hipcub::WarpReduce::TailSegmentedReduce	
cub::WarpReduce	hipcub::WarpReduce	
cub::WarpScan	hipcub::WarpScan	
cub::WarpScan::TempStorage	hipcub::WarpScan::TempStorage	
cub::WarpScan::InclusiveSum	hipcub::WarpScan::InclusiveSum	
cub::WarpScan::ExclusiveSum	hipcub::WarpScan::ExclusiveSum	
cub::WarpScan::InclusiveScan	hipcub::WarpScan::InclusiveScan	
cub::WarpScan::ExclusiveScan	hipcub::WarpScan::ExclusiveScan	
cub::WarpScan::Scan	hipcub::WarpScan::Scan	
cub::WarpScan::Broadcast	hipcub::WarpScan::Broadcast	
cub::WarpStore	hipcub::WarpStore	
cub::WarpStore::TempStorage	hipcub::WarpStore::TempStorage	
cub::WarpStore::Store	hipcub::WarpStore::Store	

5.8.2 Block-wide "collective" primitives

CUDA	HIP	CUDA version*
cub::BlockAdjacentDifference	cub::BlockAdjacentDifference	
cub::BlockAdjacentDifference::TempStorage	cub::BlockAdjacentDifference::TempStorage	
cub::BlockAdjacentDifference::BlockAdjacentDifference	cub::BlockAdjacentDifference::BlockAdjacentDifference	
cub::BlockAdjacentDifference::SubtractLeft	cub::BlockAdjacentDifference::SubtractLeft	
cub::BlockAdjacentDifference::S	cub::BlockAdjacentDifference::Su	

ubtractLeftPartialTile	btractLeftPartialTile	
cub::BlockAdjacentDifference::SubtractRight	cub::BlockAdjacentDifference::SubtractRight	
cub::BlockAdjacentDifference::SubtractRightPartialTile	cub::BlockAdjacentDifference::SubtractRightPartialTile	
cub::BlockAdjacentDifference::FlagHeads	cub::BlockAdjacentDifference::FlagHeads	
cub::BlockAdjacentDifference::FlagTails	cub::BlockAdjacentDifference::FlagTails	
cub::BlockAdjacentDifference::FlagHeadsAndTails	cub::BlockAdjacentDifference::FlagHeadsAndTails	
cub::BlockDiscontinuity	cub::BlockDiscontinuity	
cub::BlockDiscontinuity::TempStorage	cub::BlockDiscontinuity::TempStorage	
cub::BlockDiscontinuity::FlagHeads	hipcub::BlockDiscontinuity::FlagHeads	
cub::BlockDiscontinuity::FlagTails	hipcub::BlockDiscontinuity::FlagTails	
cub::BlockDiscontinuity::FlagHeadsAndTails	hipcub::BlockDiscontinuity::FlagHeadsAndTails	
cub::BlockExchange	hipcub::BlockExchange	
cub::BlockExchange::TempStorage	hipcub::BlockExchange::TempStorage	
cub::BlockExchange::StripedToBlocked	hipcub::BlockExchange::StripedToBlocked	
cub::BlockExchange::BlockedToStriped	hipcub::BlockExchange::BlockedToStriped	
cub::BlockExchange::WarpStripedToBlocked	hipcub::BlockExchange::WarpStripedToBlocked	

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cub::BlockExchange::BlockedToWarpStriped	hipcub::BlockExchange::BlockedToWarpStriped	
cub::BlockExchange::ScatterToBlocked	hipcub::BlockExchange::ScatterToBlocked	
cub::BlockExchange::ScatterToStriped	hipcub::BlockExchange::ScatterToStriped	
cub::BlockExchange::ScatterToStripedGuarded	hipcub::BlockExchange::ScatterToStripedGuarded	
cub::BlockExchange::ScatterToStripedFlagged	hipcub::BlockExchange::ScatterToStripedFlagged	
cub::BlockHistogram	hipcub::BlockHistogram	
cub::BlockHistogram::TempStorage	hipcub::BlockHistogram::TempStorage	
cub::BlockHistogram::InitHistogram	hipcub::BlockHistogram::InitHistogram	
cub::BlockHistogram::Histogram	hipcub::BlockHistogram::Histogram	
cub::BlockHistogram::Composite	hipcub::BlockHistogram::Composite	
cub::BlockLoad	hipcub::BlockLoad	
cub::BlockLoad::TempStorage	hipcub::BlockLoad::TempStorage	
cub::BlockLoad::Load	hipcub::BlockLoad::Load	
cub::BlockMergeSort	hipcub::BlockMergeSort	
cub::BlockRadixSort	hipcub::BlockRadixSort	
cub::BlockRadixSort::TempStorage	hipcub::BlockRadixSort::TempStorage	
cub::BlockRadixSort::Sort	hipcub::BlockRadixSort::Sort	
cub::BlockRadixSort::SortDescen	hipcub::BlockRadixSort::SortDesc	

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ding	ending	
cub::BlockRadixSort::SortBlockedToStriped	hipcub::BlockRadixSort::SortBlockedToStriped	
cub::BlockRadixSort::SortDescendingBlockedToStriped	hipcub::BlockRadixSort::SortDescendingBlockedToStriped	
cub::BlockReduce	hipcub::BlockReduce	
cub::BlockReduce::TempStorage	hipcub::BlockReduce::TempStorage	
cub::BlockReduce::Reduce	hipcub::BlockReduce::Reduce	
cub::BlockReduce::Sum	hipcub::BlockReduce::Sum	
cub::BlockStore	hipcub::BlockStore	
cub::BlockStore::TempStorage	hipcub::BlockStore::TempStorage	
cub::BlockStore::Store	hipcub::BlockStore::Store	
cub::BlockScan	hipcub::BlockScan	
cub::BlockScan::TempStorage	hipcub::BlockScan::TempStorage	
cub::BlockScan::ExclusiveSum	hipcub::BlockScan::ExclusiveSum	
cub::BlockScan::ExclusiveScan	hipcub::BlockScan::ExclusiveScan	
cub::BlockScan::InclusiveSum	hipcub::BlockScan::InclusiveSum	
cub::BlockScan::InclusiveScan	hipcub::BlockScan::InclusiveScan	
cub::BlockShuffle	hipcub::BlockShuffle	
cub::BlockShuffle::TempStorage	hipcub::BlockShuffle::TempStorage	
cub::BlockShuffle::Offset	hipcub::BlockShuffle::Offset	
cub::BlockShuffle::Rotate	hipcub::BlockShuffle::Rotate	
cub::BlockShuffle::Up	hipcub::BlockShuffle::Up	
cub::BlockShuffle::Down	hipcub::BlockShuffle::Down	
cub::BlockStore	hipcub::BlockStore	
cub::BlockStore::TempStorage	hipcub::BlockStore::TempStorage	

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cub::BlockStore::Store	hipcub::BlockStore::Store	

5.8.3 Device-wide primitives

CUDA	HIP	CUDA version*
cub::DeviceAdjacentDifference	hipcub::DeviceAdjacentDifference	
cub::DeviceAdjacentDifference::SubtractLeftCopy	hipcub::DeviceAdjacentDifference::SubtractLeftCopy	
cub::DeviceAdjacentDifference::SubtractLeft	hipcub::DeviceAdjacentDifference::SubtractLeft	
cub::DeviceAdjacentDifference::SubtractRightCopy	hipcub::DeviceAdjacentDifference::SubtractRightCopy	
cub::DeviceAdjacentDifference::SubtractRight	hipcub::DeviceAdjacentDifference::SubtractRight	
cub::DeviceHistogram	hipcub::DeviceHistogram	
cub::DeviceHistogram::HistogramEven	hipcub::DeviceHistogram::HistogramEven	
cub::DeviceHistogram::MultiHistogramEven	hipcub::DeviceHistogram::MultiHistogramEven	
cub::DeviceHistogram::HistogramRange	hipcub::DeviceHistogram::HistogramRange	
cub::DeviceHistogram::MultiHistogramRange	hipcub::DeviceHistogram::MultiHistogramRange	
cub::DeviceMergeSort	hipcub::DeviceMergeSort	
cub::DeviceMergeSort::SortKeys	hipcub::DeviceMergeSort::SortKeys	
cub::DeviceMergeSort::SortKeysCopy	hipcub::DeviceMergeSort::SortKeysCopy	
cub::DeviceMergeSort::StableSor	hipcub::DeviceMergeSort::StableS	

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tPairs	ortPairs	
cub::DeviceMergeSort::StableSortKeys	hipcub::DeviceMergeSort::StableSortKeys	
cub::DeviceMergeSort::SortPairsCopy	hipcub::DeviceMergeSort::SortPairsCopy	
cub::DevicePartition	hipcub::DevicePartition	
cub::DevicePartition::Flagged	hipcub::DevicePartition::Flagged	
cub::DevicePartition::If	hipcub::DevicePartition::If	
cub::DeviceRadixSort	hipcub::DeviceRadixSort	
cub::DeviceRadixSort::SortPairs	hipcub::DeviceRadixSort::SortPairs	
cub::DeviceRadixSort::SortPairsDescending	hipcub::DeviceRadixSort::SortPairsDescending	
cub::DeviceRadixSort::SortKeys	hipcub::DeviceRadixSort::SortKeys	
cub::DeviceRadixSort::SortKeysDescending	hipcub::DeviceRadixSort::SortKeysDescending	
cub::DeviceReduce	hipcub::DeviceReduce	
cub::DeviceReduce::Reduce	hipcub::DeviceReduce::Reduce	
cub::DeviceReduce::Sum	hipcub::DeviceReduce::Sum	
cub::DeviceReduce::Min	hipcub::DeviceReduce::Min	
cub::DeviceReduce::ArgMin	hipcub::DeviceReduce::ArgMin	
cub::DeviceReduce::Max	hipcub::DeviceReduce::Max	
cub::DeviceReduce::ArgMax	hipcub::DeviceReduce::ArgMax	
cub::DeviceReduce::ReduceByKey	hipcub::DeviceReduce::ReduceByKey	
cub::DeviceRunLengthEncode	hipcub::DeviceRunLengthEncode	
cub::DeviceRunLengthEncode::E	hipcub::DeviceRunLengthEncode::	

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ncode	Encode	
cub::DeviceRunLengthEncode::NonTrivialRuns	hipcub::DeviceRunLengthEncode::NonTrivialRuns	
cub::DeviceScan	hipcub::DeviceScan	
cub::DeviceScan::ExclusiveSum	hipcub::DeviceScan::ExclusiveSum	
cub::DeviceScan::ExclusiveScan	hipcub::DeviceScan::ExclusiveScan	
cub::DeviceScan::InclusiveSum	hipcub::DeviceScan::InclusiveSum	
cub::DeviceScan::InclusiveScan	hipcub::DeviceScan::InclusiveScan	
cub::DeviceScan::ExclusiveSumByKey	hipcub::DeviceScan::ExclusiveSumByKey	
cub::DeviceScan::ExclusiveScanByKey	hipcub::DeviceScan::ExclusiveScanByKey	
cub::DeviceScan::InclusiveSumByKey	hipcub::DeviceScan::InclusiveSumByKey	
cub::DeviceScan::InclusiveScanByKey	hipcub::DeviceScan::InclusiveScanByKey	
cub::DeviceSelect	hipcub::DeviceSelect	
cub::DeviceSelect::Flagged	hipcub::DeviceSelect::Flagged	
cub::DeviceSelect::If	hipcub::DeviceSelect::If	
cub::DeviceSelect::Unique	hipcub::DeviceSelect::Unique	
cub::DeviceSelect::UniqueByKey	hipcub::DeviceSelect::UniqueByKey	
cub::DeviceSpmv	hipcub::DeviceSpmv	
cub::DeviceSpmv::CsrMV	hipcub::DeviceSpmv::CsrMV	
cub::DeviceSegmentedRadixSort	hipcub::DeviceSegmentedRadixSort	

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cub::DeviceSegmentedRadixSort::SortPairs	hipcub::DeviceSegmentedRadixSort::SortPairs	
cub::DeviceSegmentedRadixSort::SortPairsDescending	hipcub::DeviceSegmentedRadixSort::SortPairsDescending	
cub::DeviceSegmentedRadixSort::SortKeys	hipcub::DeviceSegmentedRadixSort::SortKeys	
cub::DeviceSegmentedRadixSort::SortKeysDescending	hipcub::DeviceSegmentedRadixSort::SortKeysDescending	
cub::DeviceSegmentedReduce	hipcub::DeviceSegmentedReduce	
cub::DeviceSegmentedReduce::Reduce	hipcub::DeviceSegmentedReduce::Reduce	
cub::DeviceSegmentedReduce::Sum	hipcub::DeviceSegmentedReduce::Sum	
cub::DeviceSegmentedReduce::Min	hipcub::DeviceSegmentedReduce::Min	
cub::DeviceSegmentedReduce::ArgMin	hipcub::DeviceSegmentedReduce::ArgMin	
cub::DeviceSegmentedReduce::Max	hipcub::DeviceSegmentedReduce::Max	
cub::DeviceSegmentedReduce::ArgMax	hipcub::DeviceSegmentedReduce::ArgMax	
cub::DeviceSegmentedSort	hipcub::DeviceSegmentedSort	
cub::DeviceSegmentedSort::SortKeys	hipcub::DeviceSegmentedSort::SortKeys	
cub::DeviceSegmentedSort::SortKeysDescending	hipcub::DeviceSegmentedSort::SortKeysDescending	
cub::DeviceSegmentedSort::StableSortKeys	hipcub::DeviceSegmentedSort::StableSortKeys	
cub::DeviceSegmentedSort::Stabl	hipcub::DeviceSegmentedSort::Sta	

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eSortKeysDescending	bleSortKeysDescending	
cub::DeviceSegmentedSort::StableSortKeys	hipcub::DeviceSegmentedSort::StableSortKeys	
cub::DeviceSegmentedSort::StableSortKeysDescending	hipcub::DeviceSegmentedSort::StableSortKeysDescending	
cub::DeviceSegmentedSort::SortPairs	hipcub::DeviceSegmentedSort::SortPairs	
cub::DeviceSegmentedSort::SortPairsDescending	hipcub::DeviceSegmentedSort::SortPairsDescending	
cub::DeviceSegmentedSort::StableSortPairs	hipcub::DeviceSegmentedSort::StableSortPairs	
cub::DeviceSegmentedSort::StableSortPairsDescending	hipcub::DeviceSegmentedSort::StableSortPairsDescending	

5.8.4 Utilities

CUDA	HIP	CUDA version*
cub::ArgIndexInputIterator	hipcub::ArgIndexInputIterator	
cub::CacheModifiedInputIterator	hipcub::CacheModifiedInputIterator	
cub::CacheModifiedOutputIterator	hipcub::CacheModifiedOutputIterator	
cub::ConstantInputIterator	hipcub::ConstantInputIterator	
cub::DiscardOutputIterator	hipcub::DiscardOutputIterator	
cub::TexObjInputIterator	hipcub::TexObjInputIterator	
cub::TexRefInputIterator	hipcub::TexRefInputIterator	
cub::TransformInputIterator	hipcub::TransformInputIterat	

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	or	
cub::CacheLoadModifier	hipcub::CacheLoadModifier	
cub::CacheStoreModifier	hipcub::CacheStoreModifier	
cub::LoadDirectBlocked	hipcub::LoadDirectBlocked	
cub::LoadDirectBlockedVectorized	hipcub::LoadDirectBlockedVectorized	
cub::StoreDirectBlocked	hipcub::StoreDirectBlocked	
cub::StoreDirectBlockedVectorized	hipcub::StoreDirectBlockedVectorized	
cub::LoadDirectStriped	hipcub::LoadDirectStriped	
cub::StoreDirectStriped	hipcub::StoreDirectStriped	
cub::LoadDirectWarpStriped	hipcub::LoadDirectWarpStriped	
cub::StoreDirectWarpStriped	hipcub::StoreDirectWarpStriped	
cub::ThreadLoad	hipcub::ThreadLoad	
cub::ThreadStore	hipcub::ThreadStore	
cub::SHR_ADD	hipcub::SHR_ADD	
cub::BFE	hipcub::BFE	
cub::IADD3	hipcub::IADD3	
cub::PRMT	hipcub::PRMT	
cub::ThreadExit		
cub::ThreadTrap		
cub::RowMajorTid	hipcub::RowMajorTid	
cub::LaneId	hipcub::LaneId	
cub::WarpId	hipcub::WarpId	
cub::WarpMask	hipcub::WarpMask	
cub::LaneMaskLt	hipcub::LaneMaskLt	

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cub::LaneMaskLe	hipcub::LaneMaskLe	
cub::LaneMaskGt	hipcub::LaneMaskGt	
cub::LaneMaskGe	hipcub::LaneMaskGe	
cub::CachingDeviceAllocator	hipcub::CachingDeviceAllocator	
cub::SwitchDevice		
cub::KernelConfig		
cub::ChainedPolicy		
CubDebug		
CubDebugExit		
_CubLog		
cub::Debug	hipcub::Debug	
cub::CurrentDevice		
cub::DeviceCountUncached		
cub::PtxVersion		
cub::SmVersionUncached		
cub::SmVersion		
cub::SyncStream		
cub::CUB_MAX		
CUB_ALIGN		
CUB_PREVENT_MACRO_SUBSTITUTION		
CUB_MAX		
CUB_MIN		
CUB_QUOTIENT_FLOOR		
CUB_QUOTIENT_CEILING		
CUB_ROUND_UP_NEAREST		
CUB_ROUND_DOWN_NEAREST		

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CUB_STATIC_ASSERT		
CubVector		

5.8.5 GridModule

CUDA	HIP	CUDA version*
cub::GridBarrier	hipcub::GridBarrier	
cub::GridBarrier::Sync	hipcub::GridBarrier::Sync	
cub::GridBarrier::SyncFlag	hipcub::GridBarrier::SyncFlag	
cub::GridBarrierLifetime	hipcub::GridBarrierLifetime	
cub::GridBarrierLifetime::HostReset	hipcub::GridBarrierLifetime::HostReset	
cub::GridBarrierLifetime::Setup	hipcub::GridBarrierLifetime::Setup	
cub::GridBarrierLifetime::sync_bytes	hipcub::GridBarrierLifetime::sync_bytes	
cub::GridEvenShare	hipcub::GridEvenShare	
cub::GridEvenShare::DispatchInit	hipcub::GridEvenShare::DispatchInit	
cub::GridEvenShare::BlockInit	hipcub::GridEvenShare::BlockInit	
cub::GridQueue	hipcub::GridQueue	
cub::GridQueue::AllocationSize	hipcub::GridQueue::AllocationSize	
cub::GridQueue::FillAndResetDrain	hipcub::GridQueue::FillAndResetDrain	
cub::GridQueue::ResetDrain	hipcub::GridQueue::ResetDrain	
cub::GridQueue::ResetFill	hipcub::GridQueue::ResetFill	
cub::GridQueue::FillSize	hipcub::GridQueue::FillSize	

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cub::GridQueue::Drain	hipcub::GridQueue::Drain	
cub::GridQueue::Fill	hipcub::GridQueue::Fill	
cub::GridMappingStrategy	hipcub::GridMappingStrategy	
GRID_MAPPING_RAKE	GRID_MAPPING_RAKE	
GRID_MAPPING_STRIP_MINE	GRID_MAPPING_STRIP_MINE	
GRID_MAPPING_DYNAMIC	GRID_MAPPING_DYNAMIC	

5.9 thrust

5.9.1 Algorithms

CUDA	HIP	CUDA version*
thrust::copy	thrust::copy	
thrust::copy_n	thrust::copy_n	
thrust::swap_ranges	thrust::swap_ranges	
thrust::uninitialized_copy	thrust::uninitialized_copy	
thrust::uninitialized_copy_n	thrust::uninitialized_copy_n	
thrust::gather	thrust::gather	
thrust::gather_if	thrust::gather_if	
thrust::scatter	thrust::scatter	
thrust::scatter_if	thrust::scatter_if	
thrust::merge	thrust::merge	
thrust::merge_by_key	thrust::merge_by_key	
thrust::inclusive_scan	thrust::inclusive_scan	
thrust::exclusive_scan	thrust::exclusive_scan	
thrust::inclusive_scan_by_key	thrust::inclusive_scan_by_key	
thrust::exclusive_scan_by_key	thrust::exclusive_scan_by_key	
thrust::transform_inclusive_scan	thrust::transform_inclusive_scan	

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thrust::transform_exclusive_scan	thrust::transform_exclusive_scan	
thrust::reduce	thrust::reduce	
thrust::reduce_by_key	thrust::reduce_by_key	
thrust::equal	thrust::equal	
thrust::count	thrust::count	
thrust::count_if	thrust::count_if	
thrust::min_element	thrust::min_element	
thrust::max_element	thrust::max_element	
thrust::minmax_element	thrust::minmax_element	
thrust::all_of	thrust::all_of	
thrust::any_of	thrust::any_of	
thrust::none_of	thrust::none_of	
thrust::is_partitioned	thrust::is_partitioned	
thrust::is_sorted	thrust::is_sorted	
thrust::is_sorted_until	thrust::is_sorted_until	
thrust::inner_product	thrust::inner_product	
thrust::transform_reduce	thrust::transform_reduce	
thrust::partition	thrust::partition	
thrust::partition_copy	thrust::partition_copy	
thrust::stable_partition	thrust::stable_partition	
thrust::stable_partition_copy	thrust::stable_partition_copy	
thrust::shuffle	thrust::shuffle	
thrust::shuffle_copy	thrust::shuffle_copy	
thrust::copy_if	thrust::copy_if	
thrust::remove	thrust::remove	
thrust::remove_copy	thrust::remove_copy	
thrust::remove_if	thrust::remove_if	
thrust::remove_copy_if	thrust::remove_copy_if	

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thrust::unique	thrust::unique	
thrust::unique_copy	thrust::unique_copy	
thrust::unique_by_key	thrust::unique_by_key	
thrust::unique_by_key_copy	thrust::unique_by_key_copy	
thrust::unique_count	thrust::unique_count	
thrust::find	thrust::find	
thrust::find_if	thrust::find_if	
thrust::find_if_not	thrust::find_if_not	
thrust::mismatch	thrust::mismatch	
thrust::partition_point	thrust::partition_point	
thrust::set_difference	thrust::set_difference	
thrust::set_intersection	thrust::set_intersection	
thrust::set_symmetric_difference	thrust::set_symmetric_difference	
thrust::set_union	thrust::set_union	
thrust::set_difference_by_key	thrust::set_difference_by_key	
thrust::set_symmetric_difference _by_key	thrust::set_symmetric_difference _by_key	
thrust::set_union_by_key	thrust::set_union_by_key	
thrust::sort	thrust::sort	
thrust::stable_sort	thrust::stable_sort	
thrust::sort_by_key	thrust::sort_by_key	
thrust::stable_sort_by_key	thrust::stable_sort_by_key	
thrust::adjacent_difference	thrust::adjacent_difference	
thrust::generate	thrust::generate	
thrust::generate_n	thrust::generate_n	
thrust::sequence	thrust::sequence	
thrust::tabulate	thrust::tabulate	
thrust::transform	thrust::transform	

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thrust::transform_if	thrust::transform_if	
thrust::fill	thrust::fill	
thrust::fill_n	thrust::fill_n	
thrust::uninitialized_fill	thrust::uninitialized_fill	
thrust::uninitialized_fill_n	thrust::uninitialized_fill_n	
thrust::for_each	thrust::for_each	
thrust::for_each_n	thrust::for_each_n	
thrust::replace	thrust::replace	
thrust::replace_if	thrust::replace_if	
thrust::replace_copy	thrust::replace_copy	

5.9.2 Containers

CUDA	HIP	CUDA version*
thrust::host_vector	thrust::host_vector	
thrust::device_vector	thrust::device_vector	

5.9.3 Parallel Execution Policies

CUDA	HIP	CUDA version*
thrust::host_execution_policy	thrust::host_execution_policy	
thrust::device_execution_policy	thrust::device_execution_policy	
thrust::host	thrust::host	
thrust::device	thrust::device	

5.9.4 Function Objects

CUDA	HIP	CUDA version*
thrust::unary_function	thrust::unary_function	
thrust::binary_function	thrust::binary_function	

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thrust::unary_negate	thrust::unary_negate	
thrust::binary_negate	thrust::binary_negate	
thrust::zip_function	thrust::zip_function	
thrust::not1	thrust::not1	
thrust::not2	thrust::not2	
thrust::make_zip_function	thrust::make_zip_function	
thrust::placeholders	thrust::placeholders	
thrust::plus	thrust::plus	
thrust::minus	thrust::minus	
thrust::multiplies	thrust::multiplies	
thrust::divides	thrust::divides	
thrust::modulus	thrust::modulus	
thrust::negate	thrust::negate	
thrust::square	thrust::square	
THRUST_UNARY_FUNCTOR_VOID_SPECIALIZATION	THRUST_UNARY_FUNCTOR_VOID_SPECIALIZATION	
THRUST_BINARY_FUNCTOR_VOID_SPECIALIZATION	THRUST_BINARY_FUNCTOR_VOID_SPECIALIZATION	
THRUST_BINARY_FUNCTOR_VOID_SPECIALIZATION_OP	THRUST_BINARY_FUNCTOR_VOID_SPECIALIZATION_OP	
thrust::bit_and	thrust::bit_and	
thrust::bit_or	thrust::bit_or	
thrust::bit_xor	thrust::bit_xor	
thrust::equal_to	thrust::equal_to	
thrust::not_equal_to	thrust::not_equal_to	
thrust::greater	thrust::greater	
thrust::less	thrust::less	
thrust::greater_equal	thrust::greater_equal	

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thrust::less_equal	thrust::less_equal	
thrust::identity	thrust::identity	
thrust::maximum	thrust::maximum	
thrust::minimum	thrust::minimum	
thrust::project1st	thrust::project1st	
thrust::project2nd	thrust::project2nd	
thrust::logical_and	thrust::logical_and	
thrust::logical_or	thrust::logical_or	
thrust::logical_not	thrust::logical_not	

5.9.5 Iterators

CUDA	HIP	CUDA version*
thrust::input_device_iterator_tag	thrust::input_device_iterator_tag	
thrust::output_device_iterator_tag	thrust::output_device_iterator_tag	
thrust::forward_device_iterator_tag	thrust::forward_device_iterator_tag	
thrust::bidirectional_device_iterator_tag	thrust::bidirectional_device_iterator_tag	
thrust::random_access_device_iterator_tag	thrust::random_access_device_iterator_tag	
thrust::input_host_iterator_tag	thrust::input_host_iterator_tag	
thrust::output_host_iterator_tag	thrust::output_host_iterator_tag	
thrust::forward_host_iterator_tag	thrust::forward_host_iterator_tag	
thrust::bidirectional_host_iterator_tag	thrust::bidirectional_host_iterator_tag	
thrust::random_access_host_iterator_tag	thrust::random_access_host_iterator_tag	

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thrust::constant_iterator	thrust::constant_iterator	
thrust::counting_iterator	thrust::counting_iterator	
thrust::discard_iterator	thrust::discard_iterator	
thrust::iterator_adaptor	thrust::iterator_adaptor	
thrust::iterator_facade	thrust::iterator_facade	
thrust::iterator_core_access	thrust::iterator_core_access	
thrust::permutation_iterator	thrust::permutation_iterator	
thrust::reverse_iterator	thrust::reverse_iterator	
thrust::transform_input_output_iterator	thrust::transform_input_output_iterator	
thrust::transform_iterator	thrust::transform_iterator	
thrust::transform_output_iterator	thrust::transform_output_iterator	
thrust::zip_iterator	thrust::zip_iterator	
thrust::make_constant_iterator	thrust::make_constant_iterator	
thrust::make_counting_iterator	thrust::make_counting_iterator	
thrust::make_discard_iterator	thrust::make_discard_iterator	
thrust::make_permutation_iterator	thrust::make_permutation_iterator	
thrust::make_reverse_iterator	thrust::make_reverse_iterator	
thrust::make_transform_input_output_iterator	thrust::make_transform_input_output_iterator	
thrust::make_transform_iterator	thrust::make_transform_iterator	
thrust::make_transform_output_iterator	thrust::make_transform_output_iterator	
thrust::make_zip_iterator	thrust::make_zip_iterator	
thrust::advance	thrust::advance	
thrust::distance	thrust::distance	

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5.9.6 Memory Management

CUDA	HIP	CUDA version*
thrust::device_reference	thrust::device_reference	
thrust::device_ptr	thrust::device_ptr	
thrust::device_delete	thrust::device_delete	
thrust::device_free	thrust::device_free	
thrust::device_malloc	thrust::device_malloc	
thrust::device_new	thrust::device_new	
thrust::operator<<	thrust::operator<<	
thrust::device_pointer_cast	thrust::device_pointer_cast	
thrust::swap	thrust::swap	
thrust::get_temporary_buffer	thrust::get_temporary_buffer	
thrust::return_temporary_buffer	thrust::return_temporary_buffer	
thrust::raw_pointer_cast	thrust::raw_pointer_cast	
thrust::raw_reference_cast	thrust::raw_reference_cast	
thrust::device_ptr_memory_resource	thrust::device_ptr_memory_resource	
thrust::device_allocator	thrust::device_allocator	
thrust::device_malloc_allocator	thrust::device_malloc_allocator	
thrust::device_new_allocator	thrust::device_new_allocator	
thrust::mr::allocator	thrust::mr::allocator	
thrust::mr::stateless_resource_allocator	thrust::mr::stateless_resource_allocator	
thrust::mr::polymorphic_allocator	thrust::mr::polymorphic_allocator	
thrust::mr::operator==	thrust::mr::operator==	
thrust::mr::operator!=	thrust::mr::operator!=	
thrust::mr::disjoint_unsynchronize	thrust::mr::disjoint_unsynchronize	

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d_pool_resource	d_pool_resource	
thrust::mr::disjoint_synchronized_pool_resource	thrust::mr::disjoint_synchronized_pool_resource	
thrust::mr::memory_resource thrust::mr::new_delete_resource	thrust::mr::memory_resource thrust::mr::new_delete_resource	
thrust::mr::unsynchronized_pool_resource	thrust::mr::unsynchronized_pool_resource	
thrust::mr::pool_options	thrust::mr::pool_options	
thrust::mr::synchronized_pool_resource	thrust::mr::synchronized_pool_resource	
thrust::system::cpp::memory_resource	thrust::system::cpp::memory_resource	
thrust::system::cpp::universal_memory_resource	thrust::system::cpp::universal_memory_resource	
thrust::system::cpp::universal_host_pinned_memory_resource	thrust::system::cpp::universal_host_pinned_memory_resource	
thrust::system::omp::memory_resource		
thrust::system::omp::universal_memory_resource		
thrust::system::tbb::universal_host_pinned_memory_resource		
thrust::universal_ptr	thrust::universal_ptr	
thrust::mr::tls_disjoint_pool	thrust::mr::tls_disjoint_pool	
thrust::mr::get_global_resource	thrust::mr::get_global_resource	
thrust::mr::tls_pool	thrust::mr::tls_pool	

5.9.7 Complex Numbers

CUDA	HIP	CUDA version*

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thrust::complex	thrust::complex	
thrust::complex::value_type	thrust::complex::value_type	
thrust::complex::complex	thrust::complex::complex	
thrust::complex::operator=	thrust::complex::operator=	
thrust::complex::operator+=	thrust::complex::operator+=	
thrust::complex::operator-=	thrust::complex::operator-=	
thrust::complex::operator*=	thrust::complex::operator*=	
thrust::complex::operator/=	thrust::complex::operator/=	
thrust::complex::real	thrust::complex::real	
thrust::complex::imag	thrust::complex::imag	
thrust::abs	thrust::abs	
thrust::arg	thrust::arg	
thrust::norm	thrust::norm	
thrust::conj	thrust::conj	
thrust::polar	thrust::polar	
thrust::proj	thrust::proj	
thrust::operator+	thrust::operator+	
thrust::operator-	thrust::operator-	
thrust::operator*	thrust::operator*	
thrust::operator/	thrust::operator/	
thrust::exp	thrust::exp	
thrust::log	thrust::log	
thrust::log10	thrust::log10	
thrust::pow	thrust::pow	
thrust::sqrt	thrust::sqrt	
thrust::cos	thrust::cos	
thrust::sin	thrust::sin	
thrust::tan	thrust::tan	

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thrust::cosh	thrust::cosh	
thrust::sinh	thrust::sinh	
thrust::tanh	thrust::tanh	
thrust::acos	thrust::acos	
thrust::asin	thrust::asin	
thrust::atan	thrust::atan	
thrust::acosh	thrust::acosh	
thrust::asinh	thrust::asinh	
thrust::atanh	thrust::atanh	
thrust::operator<<	thrust::operator<<	
thrust::operator>>	thrust::operator>>	
thrust::operator==	thrust::operator==	
thrust::operator!=	thrust::operator!=	

5.9.8 Random Number Generation

CUDA	HIP	CUDA version*
thrust::random	thrust::random	
thrust::random::ranlux24	thrust::random::ranlux24	
thrust::random::ranlux48	thrust::random::ranlux48	
thrust::random::taus88	thrust::random::taus88	
thrust::random::default_random_engine	thrust::random::default_random_engine	
thrust::random::minstd_rand0	thrust::random::minstd_rand0	
thrust::random::minstd_rand	thrust::random::minstd_rand	
thrust::random::ranlux24_base	thrust::random::ranlux24_base	
thrust::random::ranlux48_base	thrust::random::ranlux48_base	
thrust::random::normal_distribution	thrust::random::normal_distribution	

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thrust::random::uniform_int_distribution	thrust::random::uniform_int_distribution	
thrust::random::uniform_real_distribution	thrust::random::uniform_real_distribution	
thrust::random::operator==	thrust::random::operator==	
thrust::random::operator!=	thrust::random::operator!=	
thrust::random::operator<<	thrust::random::operator<<	
thrust::random::operator>>	thrust::random::operator>>	
thrust::random::operator==	thrust::random::operator==	
thrust::random::operator!=	thrust::random::operator!=	
thrust::random::discard_block_engine	thrust::random::discard_block_engine	
thrust::random::xor_combine_engine	thrust::random::xor_combine_engine	
thrust::random::linear_congruential_engine	thrust::random::linear_congruential_engine	
thrust::random::linear_feedback_shift_engine	thrust::random::linear_feedback_shift_engine	
thrust::random::subtract_with_carry_engine	thrust::random::subtract_with_carry_engine	

5.9.9 System

CUDA	HIP	CUDA version*
thrust::cpp	thrust::cpp	
thrust::system	thrust::system	
thrust::system::omp		
thrust::system::cpp	thrust::system::cpp	
thrust::system::cuda	thrust::system::hip	

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thrust::system::tbb		
thrust::omp		
thrust::tbb		
thrust::system::cpp::pointer	thrust::system::cpp::pointer	
thrust::system::cpp::universal_pointer	thrust::system::cpp::universal_pointer	
thrust::system::cpp::reference	thrust::system::cpp::reference	
thrust::system::omp::pointer		
thrust::system::omp::universal_pointer		
thrust::system::omp::reference		
thrust::system::tbb::pointer		
thrust::system::tbb::universal_pointer		
thrust::system::tbb::reference		
thrust::system::errc	thrust::system::errc	
thrust::system::is_error_code_enum	thrust::system::is_error_code_enum	
thrust::system::is_error_condition_enum	thrust::system::is_error_condition_enum	
thrust::system::error_category	thrust::system::error_category	
thrust::system::error_code	thrust::system::error_code	
thrust::system::error_condition	thrust::system::error_condition	
thrust::system::system_error	thrust::system::system_error	
thrust::system::generic_category	thrust::system::generic_category	
thrust::system::make_error_code	thrust::system::make_error_code	
thrust::system::operator<	thrust::system::operator<	
thrust::system::operator<<	thrust::system::operator<<	

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thrust::system::make_error_condition	thrust::system::make_error_condition	
thrust::system::operator==	thrust::system::operator==	
thrust::system::operator==	thrust::system::operator==	
thrust::system::operator!=	thrust::system::operator!=	
thrust::system::system_error::code	thrust::system::system_error::code	
thrust::system::system_error::what	thrust::system::system_error::what	

5.9.10 Utility

CUDA	HIP	CUDA version*
thrust::pair	thrust::pair	
thrust::tuple	thrust::tuple	
thrust::tuple_element	thrust::tuple_element	
thrust::tuple_size	thrust::tuple_size	
thrust::swap	thrust::swap	
thrust::make_pair	thrust::make_pair	
thrust::get	thrust::get	
thrust::make_tuple	thrust::make_tuple	
thrust::tie	thrust::tie	
thrust::proclaim_contiguous_iterator	thrust::proclaim_contiguous_iterator	
thrust::conjunction_value	thrust::conjunction_value	
thrust::disjunction_value	thrust::disjunction_value	
thrust::negation_value	thrust::negation_value	
thrust::remove_cvref	thrust::remove_cvref	
thrust::voider	thrust::voider	

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thrust::integer_sequence	thrust::integer_sequence	
thrust::index_sequence	thrust::index_sequence	
thrust::make_integer_sequence	thrust::make_integer_sequence	
thrust::make_index_sequence	thrust::make_index_sequence	
thrust::make_reversed_integer_sequence	thrust::make_reversed_integer_sequence	
thrust::make_reversed_index_sequence	thrust::make_reversed_index_sequence	
thrust::integer_sequence_push_front	thrust::integer_sequence_push_front	
thrust::integer_sequence_push_back	thrust::integer_sequence_push_back	
thrust::is_contiguous_iterator	thrust::is_contiguous_iterator	
thrust::is_execution_policy	thrust::is_execution_policy	
thrust::is_operator_less_function_object	thrust::is_operator_less_function_object	
thrust::is_operator_greater_function_object	thrust::is_operator_greater_function_object	
thrust::is_operator_less_or_greater_function_object	thrust::is_operator_less_or_greater_function_object	
thrust::is_operator_plus_function_object	thrust::is_operator_plus_function_object	
thrust::conjunction	thrust::conjunction	
thrust::conjunction_v	thrust::conjunction_v	
thrust::disjunction	thrust::disjunction	
thrust::disjunction_v	thrust::disjunction_v	
thrust::negation	thrust::negation	
thrust::negation_v	thrust::negation_v	

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thrust::remove_cvref_t	thrust::remove_cvref_t	
thrust::is_contiguous_iterator_v	thrust::is_contiguous_iterator_v	
thrust::is_execution_policy_v	thrust::is_execution_policy_v	
thrust::is_operator_less_function_object_v	thrust::is_operator_less_function_object_v	
thrust::is_operator_greater_function_object_v	thrust::is_operator_greater_function_object_v	
thrust::is_operator_less_or_greater_function_object_v	thrust::is_operator_less_or_greater_function_object_v	
thrust::system::system_error::what	thrust::system::system_error::what	
thrust::is_operator_plus_function_object_v	thrust::is_operator_plus_function_object_v	
thrust::conjunction_value_v	thrust::conjunction_value_v	
thrust::disjunction_value_v	thrust::disjunction_value_v	
thrust::negation_value_v	thrust::negation_value_v	
THRUST_PROCLAIM_CONTIGUOUS_ITERATOR	THRUST_PROCLAIM_CONTIGUOUS_ITERATOR	

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6 通讯组件兼容性列表

6.1 RCCL

6.1.1 通讯组创建及管理类函数

NCCL	RCCL	NCCL version
ncclGetLastError	ncclGetLastError	
ncclGetVersion	ncclGetVersion	
ncclGetUniqueId	ncclGetUniqueId	
ncclCommInitRank	ncclCommInitRank	
ncclCommInitAll	ncclCommInitAll	
ncclCommInitRankConfig		2.14
ncclCommFinalize		2.14
ncclCommDestroy	ncclCommDestroy	
ncclCommAbort	ncclCommAbort	
ncclCommGetAsyncError	ncclCommGetAsyncError	
ncclCommCount	ncclCommCount	
ncclCommCuDevice	ncclCommCuDevice	
ncclCommUserRank	ncclCommUserRank	

6.1.2 集合通讯函数

NCCL	RCCL	NCCL version
ncclAllReduce	ncclAllReduce	
ncclBroadcast	ncclBroadcast	
ncclReduce	ncclReduce	
ncclAllGather	ncclAllGather	
ncclReduceScatter	ncclReduceScatter	

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6.1.3 组调用函数

NCCL	RCCL	NCCL version
ncclGroupStart	ncclGroupStart	
ncclGroupEnd	ncclGroupEnd	

6.1.4 点对点通讯函数

NCCL	RCCL	NCCL version
ncclSend	ncclSend	
ncclRecv	ncclRecv	

6.1.5 数据类型

NCCL	RCCL	NCCL version
ncclComm_t	ncclComm_t	
ncclResult_t	ncclResult_t	
ncclDataType_t	ncclDataType_t	
ncclRedOp_t	ncclRedOp_t	
ncclScalarResidence_t	ncclScalarResidence_t	
ncclConfig_t		2.14

6.1.6 自定义规约操作符

NCCL	RCCL	NCCL version
ncclRedOpCreatePreMulSum	ncclRedOpCreatePreMulSum	
ncclRedOpDestroy	ncclRedOpDestroy	

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